

Candidate readings of the Xiongnu record

Forty readings of Chinese-transcribed Xiongnu names and titles as Turkic words, with the historical parallels that suggested each one. Each reading priced against the measurements in the paper. Supplied for the record; not offered as results.

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Supplement to *Reading steppe names: what a transcription channel can and cannot recover from the Chinese record of the Xiongnu*

S1 · author_proposals.csv · compiled 2026

S1.1 What this supplement is

Why the file is worth having, given that none of it is a finding. A reading of a Xiongnu name has been offered many times over a century, and offered as a resemblance: this Chinese spelling looks like that Turkic word. What is new here is not the readings but that each one is *priced*. Every row sets out what its reading would require a scribe to have done, and gives the rate at which the corpora show scribes doing it, so that a reader who doubts a reading can see which step to attack and what it would cost to defend. The same treatment is applied to proposals that are not ours, and it is applied to this author without exemption: several readings here were withdrawn while the file was being written because a measurement ruled them out, and §S1.14 records which. A reading that survives that is still not a finding. But it is a different object from a resemblance, and that difference is what the file is for.

Section 9 of the paper sets out what a proposed reading of a Xiongnu name has to survive before it can be called a finding: the sense has to be attested in a source independent of the comparison, the morphology has to be licensed by a suffix with the right distribution, and any objection that can be tested has to be tested. The file covers all forty items of the record, and they are not all of one kind. **One** reading is argued in the paper itself and meets its standard: 若鞮 as *Īnakt* (§9). **One** is the standard identification, not ours and not in dispute, and serves as this file's calibration control: 撑犁 as *Tanrı*, the only two-character item whose meaning the Chinese sources state outright. **Two** carry comparisons taken from the literature rather than made by us, which the paper then tests: 徑路 as *Kınrak* and 屠耆 as *Doğru* (§9.1). **One** is a compound pointing at its constituent rows. Every row now carries a reading; none is left blank. The remaining **thirty-five** are the author's own proposals and none of them meets the standard, 冒頓 among them: §9.2 of the paper measures the three values its second character can take and does not choose, and the proposal made here is set out in S1.4 with its objections.

They are here since we produced them over the course of the work. A reader who has seen 稽粥 written out as *kei çuk* will hear Turkish *küçük* whether or not this supplement exists. What can be done is to write them down, date them, label them, and put next to each one the reason it occurred to us, so that the reasoning is inspectable rather than implied.

Nothing in the paper depends on this supplement. The measurements in §§5–8 (the channel accuracies, the period-gap decomposition, the initial-consonant probe, the coda-spelling count) were made without reference to these readings and do not change if every one of them is rejected.

A dagger, †, on a syllable in the tables below marks a **polyphonic character**, one for which Schuessler's table gives more than one Later Han reading. The reading printed is the first tabulated one, as in Appendix A of the paper. Table S7 lists every such character with the readings not used, and where an alternative would change a proposed reading the entry for that item says so.

S1.2 Why they are not in the paper

Section 6.3 measures exactly the operation that produced this supplement, taking a reconstructed reading and looking it up in a Turkic dictionary, against a null model built by resampling Later Han syllables into pseudo-names. Across seven lexicons, from modern Turkish to the *Codex Cumanicus*, between none and three names in thirty-six meet significance of $p \leq 0.05$. At a one-in-twenty threshold, thirty-six coin flips give the same.

Doing the matching by hand does not improve on the algorithm. It makes it worse. A person cannot help preferring the reading that means something, and that preference is precisely the selection effect the null exists to quantify. The algorithm at least fails honestly. A paper that measured the unreliability of this operation and then presented its own outputs as discoveries would be incoherent.

So we keep the two things apart. The paper reports what can be measured. This file records what we imagined while measuring it. Because it is supplementary, it does not carry the paper's evidential standard and should not be read as though it did.

S1.3 How the readings were produced

Every row starts from the same two published inputs, neither of them model output. The Chinese form is from the *Shiji*, *Hanshu* and *Hou Hanshu*. The Later Han reading is Schuessler's. The third column simply respells that reading in Turkish orthography, so that a reader who does not read phonetic notation can see what is being claimed. It adds no information.

The fourth column is ours. We arrive at it by hearing the third column as a Turkic word or compound, usually with a vowel changed, sometimes treating a consonant as lost, and then looking for a historical parallel that would make such a name plausible for a steppe ruler. The fifth column gives the sense that reading would carry. The note beneath each row gives the parallel. Each entry below is set out in two parts: *Phonetics*, which walks the Later Han reading to the proposed name character by character and says which of the steps are ordinary correspondences and which are not; and *Reason*, which gives the word, its attestation where it has one, and the naming parallel that suggested it.

The standard we used for inclusion here is deliberately lower than the paper's, and it is worth stating. A reading is offered when an attested word fits the transcription and nothing in the transcription stands against it. It is not offered when the spelling rules it out, when a character would have to carry a sound the corpora never show it carrying. Under-determination is not a reason to withhold a reading, because almost every short name in the record is under-determined; a demonstrable phonological conflict is a reason.

We had three reasoning habits. We list them because they are the main weakness. The first is the epithet argument: later Turkic rulers are commonly named by a common noun or a natural force: Bayezid I *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed IV *Avcı* "the hunter", so a reading that yields such a noun is treated as more likely. The second is the career argument: where the Chinese record says what a ruler did, a reading that names the deed is preferred. The third is the compound argument: a long transcription is split into two or three Turkic elements. Each of these makes a reading easier to reach, and none of them makes it more likely to be right. They multiply the search space, which is the opposite of evidence.

s1.4 The reading argued in the paper, and this author's proposal for 冒頓

Two items in this supplement are set out at length. The first is argued in the paper rather than merely recorded, because it is supported differently from everything else in the file: the recurring element 若鞮 *riak* *te*, which closes the formal names of six later chanyu, read *İnak*, "the faithful ones" (§9):

- the *Hou Hanshu* glosses 若鞮 as 孝 "filial", the meaning is recorded in the source, not inferred from a resemblance;
- *inak* is attested as *fidelis* "faithful" in the *Codex Cumanicus* (Kuun 1880, p. 182), a medieval source with no connection to the Chinese record;
- the morphology is licensed: *-t* is an Old Turkic plural used with titles (*tegin* → *tegit*, *tarqan* → *tarqat*), and the competing *-tI* reading we ruled out by counting: of 31 Cuman headwords in *-tI*, almost all are past-tense verbs;
- we tested the obvious objection to the spelling and it failed: if the source ended in *-t*, why did the scribe use the open character 鞮? Because 30% of stop-final source syllables in the Sanskrit corpus were written with an open character.

One objection stands and is stated in the paper: if the source word was *inak*, the initial *ɪ-* is not written, and 若鞮 has no room for it.

The second is not a reading the paper argues. 冒頓 *mək tuən^c* is the founder's name, and §9.2 prices the three values its second character can take without choosing among them. **This author's proposal is *Bögü Tuğ***, "sage standard", and it is set out here rather than there, with the objections to it, because that is where proposals in this work belong.

- **The frame.** Both codas are written, which makes it one of the tightest frames in the record, but neither character has a single value. 冒 has two Later Han readings, *mək* and *mou^c*; 頓 has *tuən^c* and *tuət*, and a third value from the commentarial note 頓音毒, which Pulleyblank reports as Early Middle Chinese *dawk*. That third value is a substitution recorded by a commentator, not a row of the table, and the paper labels it as such.
- **The first element.** 冒 and 默 share the reading *mək*, and 默噶 is read *Bögü-Çor*, the personal name of Qapaqhan Qaghan; 牟 *mu* writes the same word in 牟羽 for the Uyghur *Bögü* Qaghan. So both rows of 冒 have a precedent for writing *bögü*. Table 9 of the paper prices them: a rounded source vowel is written with a rounded Chinese vowel in 59% of cases and an open one in 4%, so the second row is where *bögü* is ordinary rather than merely permitted.
- **The second element.** *tuğ* is the horse-tail or yak-tail standard carried on a pole, one of the oldest continuously used objects of steppe rulership. It is attested in Old Turkic: *tuğ taşıkar* "when the standard went out" in the Moyun Çor inscription, cited by Doerfer at ETY I 170, and Kāšgarī glosses it as the drum and banner borne before the king, with *toquz tuğluğ xan* "a khan of nine standards". It ends in a velar, so it needs 頓's second row (2 of 21 opportunities) or the substitution (10 of 21), and it is excluded on the first row (0 of 21).
- **The count that follows.** Two of the three values of 頓 admit a velar-final second element, one admits the liquid-final *Bayatur* the field has read for a century, and one admits a nasal-final one. That is the whole of the arithmetic, and it is why this author prefers a velar-final reading.
- **An observation about the substitution.** In the Later Han table 毒 and 𪛗 are exact homophones, *douk*, and 𪛗 is the Chinese word for the yak-tail standard itself. So the commentator who wrote 頓音毒 put the value of 頓 at precisely the Later Han reading of the word for the standard. This may be coincidence, and one homophone pair proves nothing, but it is recorded here rather than left out.
- **The name shape.** *bögü* is a regnal epithet across the record: *Teŋri Bögü Teŋriken* in the Old Turkic inscriptions, *Bögü* Qaghan of the Uyghurs (牟羽), *Bögü-Çor* of the Türks (默噶). Kāšgarī glosses *bügü* as "bilgin, akıllı, hakim"; Clauson says the word "seems to connote both wisdom and mysterious spiritual power", and it was borrowed early into Mongolian as *böge*, the ordinary word for a shaman. Two nouns in apposition is the commonest Turkic name pattern.

The objections, all of which stand.

- **The loan direction.** Doerfer's *TMEN* gives Turkic *tuğ* as a loan from Chinese (entry 969). If that is right, the word cannot be an element of a name of the early second century BC, and the reading fails. Doerfer's own addendum in volume IV weakens it: it quotes Menges (1968: 169, as cited there) that the Chinese word "might well be an ancient loanword in Chinese", and H. Franke, in a letter of 23 March 1966, that the Chinese word is first attested in the *Zhouli* in a purely Chinese context and is "kaum jünger als 2. Jh. v. Chr.", which is the founder's own century. The direction is therefore disputed by the same authority usually cited for it, and this author does not settle it.
- **The row it needs.** The reading is ordinary on the substitution, marginal on 頓's second row, and excluded on the first, which is the row Appendix A of the paper prints. It also reads better on the second row of 冒 than on the first, which Appendix A likewise prints.
- **The compound is not attested.** Only its two halves are, as with every compound reading in this supplement.
- **The frame does not fix the vowel** of the second syllable, so *tuğ* is what the velar coda and the Turkic lexicon together suggest, not what the transcription requires.

The rival reading is not free of these either, and it is fair to say so. Clauson's entry on *bayatur* records that the word "occurs only once in the early period and then still as a P.N.", that the sense "picked warrior" most probably developed in Mongolian rather than Turkic, and that Turkic had other words such as *alpayut* in that meaning; Doerfer's entry 817 adds that the form *bayatur* is not findable in the Old Turkic dictionaries and texts available to him, the attested Old Turkic form being *bātur*, and that the reconstruction **bayatur* is secured by Old Hungarian *bahatur* rather than by a Turkic attestation. A form without a medial velar leaves 冒's written coda with nothing to write.

Both appear in the tables below in their proper places, so that this supplement can be read as one list with the standard visible in it. The second appears there as an author's proposal, which is what it is.

S1.5 Why the decoder's output is not cited below

The channel of §4 has a decoding step that will return a ranked list of readings for any string in the record. None of those rankings is quoted in this supplement, and the reason is worth stating, because the numbers look more informative than they are.

A character can be *reading-anchored*, meaning its emission distribution was learned for that character specifically, or it can fall back to a table keyed on the Later Han **initial consonant class alone**, discarding the vowel and the coda. Of the 39 items in the record, only five are reading-anchored throughout; 34 use backoff on at least one character and nine use it on every character. The effect is that whole groups of characters share one distribution:

LATER HAN INITIAL CLASS	CHARACTERS IN THE XIONGNU RECORD SHARING ONE EMISSION DISTRIBUTION	WHAT THE BACKOFF TABLE RETURNS FOR EVERY ONE OF THEM
velar	17: 稽 渠 胡 湖 壺 戶 侯 姑 孤 骨 谷 韓 諧 皋 權 胸 徑	ka 0.34, ga 0.15, ku 0.08
affricate	6: 且 粥 珠 臣 詹 郅	ca 0.18, ja 0.10, cā 0.08
dental stop	9: 鞮 題 當 屠 撐 株 稚 脫 頓	ta 0.23, da 0.11, ti 0.10
liquid	6: 廬 孿 祿 累 蠡 閭	ra 0.37, la 0.17, rī 0.08

TABLE S1 Because 稽 *k^hei^a* and 渠 *gia* draw on the same row, and 粥 *tśuk* and 且 *tśa^a* on another, an affricate-plus-velar string and a velar-plus-affricate string return mirror images of one table entry rather than anything specific to those characters. Two further limits apply: the scores are normalised over a retained beam, so they are relative rankings within it and not posteriors; and the syllable inventory is Sanskrit-derived, so *q* and the front rounded vowels *ō* and *ū* cannot be produced at all, so a reading containing *ū* is therefore not rejected by the decoder, it is unavailable to it. §6.2 of the paper makes the general point; this table is its mechanism.

In the entries below, where a spelling question arises, we cite the paper's initial-correspondence counts (§8.1): direct counts of which Chinese initial was used for which source sound across the three corpora. Those are observations rather than model output, and they are what the word "permitted" means wherever it appears in this supplement.

S1.6 Chanyu, in order of reign

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
頭曼 Touman	do man ^{c†}	do man	Duman	dust cloud
PHONETICS 頭 <i>do</i> → <i>du</i> : the initial is direct; o and u are not separable at the ~50% vowel recovery §7 reports. 曼 <i>man</i> → <i>man</i> : exact, and the coda -n is written, because -n is one of the six codas the Chinese syllable can carry. One vowel change in two syllables. The rival reading <i>tūmen</i> needs <i>ū</i> , which the Later Han script cannot write at all.				
REASON <i>duman</i> / <i>tuman</i> / <i>touman</i> "smoke, fog, mist" is one word across Kāšgārī, the Codex Cumanicus and modern Turkish. The standard identification is <i>tūmen</i> "ten thousand", which is absent from the Cuman vocabulary entirely.				
冒頓 Modu / Modun	mək [†] tuən ^{c†}	mık tuın	Bögü Tuğ	sage standard

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS The frame is b/m – k/g – t/d – nasal or stop, and both codas are written, which makes it one of the tightest frames in the record and also one of the least determinate, because neither character has a single value. 冒 has two Later Han readings, <i>mək</i> and <i>mou</i>; 頓 has <i>tuən</i> and <i>tuət</i>, and a third value from the commentarial note 頓音毒, reported by Pulleyblank as Early Middle Chinese <i>dawk</i>, which is a substitution recorded by a commentator rather than a row of the table. 冒 <i>mək</i> → <i>bögü</i>: the m- stands for a source b-, 3 of 22 in the Later Han corpus, 13.6%; the written -k stands for the source g; and the final vowel is not written, one character covering two source syllables, a device found in 47 of the 1,110 verified pairs, as 竭 covers <i>-gata</i> in <i>tathâgata</i>. There is a character-level precedent as well as a rate: 冒 and 默 share the reading <i>mək</i> and 默噉 is read <i>Bögü-Çor</i>, while 牟 <i>mu</i> writes the same word in 牟羽 for the Uyghur Bögü Qaghan. 頓 → <i>tuğ</i>: a velar-final second element, which the corpus writes with the -t coda of 頓’s second row in 2 of 21 opportunities and with the -k of the substitution in 11 of 23, and never with the -n of the first row, 0 of 21. REASON Bögü Tuğ, “sage standard”. <i>tuğ</i> is the horse-tail or yak-tail standard carried on a pole, attested in Old Turkic (<i>tuğ taşırkar</i> “when the standard went out”, Moyun Çor, cited by Doerfer at ETY I 170) and glossed by Kâşgarî as the drum and banner borne before the king, with <i>toquz tuğluğ xan</i> “a khan of nine standards”. <i>bögü</i> is a regnal epithet across the record: <i>Tengri Bögü Tengriken</i>, Bögü Qaghan of the Uyghurs, Bögü-Çor of the Türks; Kâşgarî glosses <i>bügü</i> as “bilgin, akıllı, hakim” and Clauson says it “seems to connote both wisdom and mysterious spiritual power”. Both halves are attested independently of any Chinese source and two nouns in apposition is the commonest Turkic name pattern. Why a velar-final second element. Of the three values 頓 can take, two admit a velar-final element, one admits the liquid-final <i>Bayatur</i> the field has read for a century, and one admits a nasal-final one. §9.2 of the paper sets out that comparison and does not choose; the preference here is this author’s. What is not settled. The loan direction of <i>tuğ</i> is disputed: Doerfer’s <i>TMEN</i> 969 gives it as a loan from Chinese, which would put it out of reach of a name of the early second century BC, while his own addendum in volume IV quotes Menges (1968: 169, as cited there) that the Chinese word “might well be an ancient loanword in Chinese” and Franke (letter of 23 March 1966) that the Chinese word is first attested in the <i>Zhouli</i> and is “kaum jünger als 2. Jh. v. Chr.”. A ninth-century check adds nothing either way, and is recorded so that it is not made twice: Tekin’s glossary to the <i>İrk Bitig</i> has the verb <i>tug-</i> “to rise (of sun)” but not the noun, the text’s word for a flag is <i>uçrug</i>, and <i>bögü</i> is not in it. That text is sixty-five omens and a few hundred words, so absence from it is not evidence of anything. The reading is also ordinary on the substitution, marginal on 頓’s second row and excluded on the first, which is the row printed here; the compound is not itself attested, only its halves; and the frame does not fix the vowel of the second syllable. The rival reading carries its own difficulties, recorded in S1.4. Full discussion in S1.4.				
稽粥 Jizhou / Jiyou	k ^h ei ^ɐ † tśuk	kei çuk	Kiçük	the younger
PHONETICS 稽 <i>k^hei</i> → <i>ki</i>: an aspirated velar is the ordinary vehicle for a source k, and the diphthong <i>ei</i> for <i>i</i> is routine. 粥 <i>tśuk</i> → <i>çük</i>: tś- is the default character class for a source affricate (37% in the Sanskrit corpus), and the coda -k is actually written, the strongest part of the reading, since a stop coda goes unwritten 30% of the time. Only ü is unavailable and surfaces as u. REASON <i>kiçig</i> / <i>kiçik</i> / <i>kiçük</i> “small, younger” is attested three times independently: Kâşgarî has <i>kiçik</i> and <i>kiçük</i> with the derivative <i>kiçiglik</i>, the <i>Kutadgu Bilig</i> has <i>kiçig oğlan erken</i> “when he was a small boy” (1823), and the Codex has <i>Kečak</i> glossed <i>paucus</i>. 稽粥 is the personal name of Laoshang chanyu, Modu’s son and successor, so “the younger” fits the succession the Chinese sources record.				
軍臣 Junchen	kun džin †	kun cin	Köngen	the one who is always upright
PHONETICS 軍 <i>kun</i> → <i>kön</i>: three steps, all direct. A source velar written with a Chinese <i>k-</i> is 204 of 236, 86.4%; a source rounded vowel written rounded is 207 of 324, 63.9%; the <i>-n</i> is written, 94 of 111, 84.7%. 臣 → <i>-gen</i> takes the second of the character’s two Later Han readings, the velar <i>gin</i>, rather than the palatal <i>džin</i> that Appendix A prints: a source <i>g</i> written with a Chinese <i>g-</i> is 64 of 110, 58.2%, and a source front vowel written with a front vowel is 349 of 406, 86.0%. The palatal reading is not available to this name. A source voiceless affricate written with the voiced <i>dž-</i> is 0 of 53 in the Sanskrit corpus, so <i>cin</i>, <i>čen</i> and <i>čan</i> are excluded outright; the Codex forms that appear to offer an affricate are an artefact of medieval Latin spelling, in which <i>j</i> writes Turkic <i>y-</i> and <i>c</i> before a back vowel writes <i>k-</i>. Step 47 prices every combination the two characters permit, and <i>köngen</i> is the likeliest at 23.4%, about 1 in 4, against 1 in 9 for the best branch of 冒頓 and 1 in 152 for the control at 撐犁. Two syllables, two characters, nothing inserted and nothing discarded. REASON Köngen, “the one who is always upright”. The stem is the verb <i>kön-</i>, which Kâşgarî glosses <i>düzelmek</i>, <i>doğrultmak</i>; <i>yola gelmek</i>, to straighten, to become upright, to come to the right way, and which carries a family around it: <i>könitmek</i>, <i>köndgermek</i> and <i>köndgürmek</i>, all <i>doğrultmak</i>. The suffix is <i>-gAn</i>, the deverbal participle with a habitual force. The suffix is counted here, not assumed. Across the 9,379 headwords of the Kâşgarî index, the Codex Cumanicus and Tekin’s glossary to the <i>İrk Bitig</i>, <i>-gAn</i> has 211 headwords and 109 of them are glossed with <i>daima</i>, <i>her zaman</i> or <i>çok</i>: <i>arıtgan</i> “her zaman temizleyen”, <i>egilgen</i> “daima eğilen”, <i>etilgen</i> “her zaman düzelen”, which is the near sibling of the form proposed here. The competing suffixes <i>-gUn</i> and <i>-gIn</i> have 72 headwords between them and not one habitual gloss: they are concrete nouns, birds and plants and body parts, a knot, a bridle. In the eleventh-century record <i>-gAn</i> is the productive participle and <i>-gUn</i> is not. The modern <i>durgun</i>, <i>çoşkun</i> and <i>ötgün</i> belong to a later productivity of <i>-gUn</i> and are not evidence for this period. Vowel harmony decides a tie the transcription cannot. <i>Köngin</i> costs exactly what <i>köngen</i> costs, 1 in 4, because 臣 carries the front vowel either way. But <i>-gAn</i> has two shapes while <i>-gin</i> and <i>-gUn</i> have four, so a front rounded stem such as <i>kön-</i> must take <i>-gün</i> and cannot take <i>-gin</i>. The Turkic evidence removes a candidate that the Chinese evidence cannot. What is not settled. <i>Köngen</i> is not itself an attested headword. It is a formation by a productive rule from an attested stem, which is a stronger position than a form requiring a segment that must afterwards be lost, but it is not an attestation. The lexicons are eleventh century and the name is second century BCE, so that <i>-gAn</i> was productive for Kâşgarî does not establish it 1250 years earlier. Whether <i>-gAn</i> formations were used as personal names or titles in Old Turkic is not tested here. The <i>g</i> to <i>g</i> rate is measured over all positions, and whether a <i>g</i> after a nasal behaves the same has not been measured. And the first element is undetermined: <i>kön</i>, <i>kün</i>, <i>kon</i> and <i>kun</i> all cost the same 47.0%, so choosing <i>kön-</i> is a choice made on meaning, which is the first of the three reasoning habits this supplement lists against itself.				
詹師廬 Zhanshilu	tśam ʃi lia	çam ʃi lia	Çamşılı	dazzling, radiant
PHONETICS 詹 <i>tśam</i> → <i>çam</i>: tś- is the default affricate spelling and the -m coda is written. 師 <i>ʃi</i> → <i>ʃr</i>: the retroflex sibilant is the standard vehicle for ʃ. 廬 <i>lia</i> → <i>lr</i>: note that Chinese l- also carries source r-: in the paper’s correspondence counts an r- was written l- in 12 of 14 attestations, so this syllable is genuinely ambiguous between -li and -ri. REASON A stem of shining or dazzling. The solar reading this once echoed at 軍臣 has been withdrawn, so it stands alone.				
兜犁湖 Xulihu / Goulihu	huo † lei † ga	huo lei ga	Kalık	sky, heaven

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 响 <i>huo</i> → <i>ka</i> , 犁 <i>lei</i> → <i>li</i> , 湖 <i>ga</i> → <i>k</i> : three onsets, each taken from the majority correspondence, with nothing inserted and no coda discarded. The velar class covers h-, k- and g- alike, so the first and third syllables cost nothing.				
REASON Kāşgarî glosses <i>kalık</i> “hava, gök, sema” (air, sky, heaven). The word belongs to the same semantic field as 撐犁, the one Xiongnu term the Chinese sources gloss (天 “heaven”), so a ruler bearing it would not be isolated in the record. A second exact fit, <i>kılık</i> “disposition, manner”, is available if the sky reading is thought too convenient.				
伊稚斜	ʔi ɖi˦ ʃa	’i di ya	İdiya	master
Yizhixie				
PHONETICS 伊 ʔi → <i>i</i> : a glottal onset is the standard way to write a vowel-initial syllable. 稚 ɖi → <i>di</i> : retroflex ɖ for d is routine. 斜 ʃa → <i>ya</i> : this character class emits <i>ya</i> more strongly than any other single-syllable correspondence in the record. All three steps are default.				
REASON <i>idi</i> “lord, master”, attested in Old Turkic. Compare Murad I's epithet <i>Hüdavendigâr</i> , “the sovereign”, a title of the same shape and force.				
烏維	ʔa wi	’a vi	Abı	one who hunts or stalks
Wuwei				
PHONETICS 烏 ʔa → <i>a</i> : vowel-initial, direct. 維 wi → <i>bi</i> is the weak step: the Chinese character carries a w-, and while Turkic b- and w- alternate, the transcription reads more naturally as <i>avı</i> than <i>abı</i> . This is one of only five items in the record decoded entirely from character-specific evidence, so the mismatch cannot be attributed to a gap in the data.				
REASON <i>av</i> “hunt”, <i>avcı</i> “hunter”. <i>Avcı</i> names an occupation; <i>abı</i> is proposed as naming a trait. <i>Avcı</i> is also the epithet of Mehmed IV.				
且鞮侯	tʂa ^B te [†] go	tʂa te go	Çatıgka	one ready for the clash
Qiedihou				
PHONETICS 且 <i>tʂa</i> → <i>ça</i> : <i>ts-</i> for a source affricate is a minority spelling but a real one, 15% in the Sanskrit corpus and 9% in Ligeti. 鞮 <i>te</i> → <i>tr</i> : direct. 侯 <i>go</i> → <i>gka</i> asks a single Chinese syllable to carry a consonant cluster it structurally cannot, so the reading adds material the transcription does not have. The price says the same thing in a number. The whole reading costs 1 in 219,014 , which is 1,441 times the control at 撐犁 and the most expensive of the proposals in this file. \$8.7 of the paper treats <i>Bayatur</i> , at 650 times the control, as a reading the method contradicts; this one is more than twice as far out, and a reader is entitled to weigh it on that scale. One way out was tested and is closed. 侯 is an ordinary Chinese noble rank, and the same three chapters use it that way throughout for Han enfeoffments, 浞野侯, 博望侯, 合騎侯, 長平侯, 重合侯. If it were the title here, the Xiongnu name would be 且鞮 alone and this character would need no reading at all. But 侯 stands medially in the Southern Xiongnu regnal titles 醯儻尸逐侯鞮, 胡邪尸逐侯鞮, 休蘭尸逐侯鞮 and 亭獨尸逐侯鞮, between 尸逐 and 鞮, where a Chinese rank cannot stand. So 侯 is transcriptional in this class of name and the third character has to be read. Nor does the lexicon offer anything better. One hundred three-syllable headwords fit all three characters and the best of them prices at 1 in 2,070 and means a snotty child.				
REASON <i>çat-</i> “to strike together, to clash”. Compare Selçuk Bey, whose name has been read as “perpetually ready for the fight”. What is not settled. Everything above. This is an author's proposal in the sense §S1 sets out, a reading the record does not forbid outright, and it is kept here so the file can be read as one list; it is not offered as supported, and on the scale §8.6 supplies it is the weakest thing in the file.				
狐鹿姑	ɣua lok ka	ğua lok ka	Kurluga	the binder, the encloser
Hulugu				
PHONETICS 狐 <i>ɣua</i> → <i>kur</i> : a velar fricative for a Turkic back <i>q</i> , which §8.5 of the paper measures at 43 of 66, 65.2%, against 20 of 66 for an aspirated character; <i>kurluga</i> has back vowels throughout. The rounded vowel is written rounded, 207 of 324, 63.9%. The <i>-r</i> is not written, and that is the ordinary treatment rather than a difficulty : Table 7 gives a source liquid coda spelled with an open character 21 times in 37, 56.8%, the majority option, and 狐 is open. 鹿 <i>lok</i> → <i>lu</i> : a source <i>l</i> written with a Chinese <i>l</i> - is 97 of 100, 97.0%, the vowel rounded again. Its written -k is not discarded. The character after it is 姑 <i>ka</i> , a velar, and a medial written coda writes the onset of the following syllable in 165 of 446 cases, 37.0%, which is the largest single thing such a coda does: 僧 for <i>saṃ-ghā-</i> , 三 for <i>saṃ-</i> , 怛 for <i>ta-thā-</i> . 姑 <i>ka</i> → <i>-ga</i> , with the vowel open, 1127 of 1634, 69.0%. The whole name costs 2.00%, about 1 in 50 , against the 1 in 152 that the control at 撐犁 gives for a reading the Chinese sources gloss outright. The weak step is the last one, not the first two. A source <i>g</i> written with a Chinese <i>k</i> - is 23 of 110, 20.9%. Were the word <i>kurluka</i> the same character would be writing a source <i>k</i> at 86.4% and the name would cost 1 in 12, so one consonant moves the price fivefold. Step 51 sets this out.				
REASON <i>Kurluga</i> , from the sense of binding and enclosing. The root is attested and so is the sense: Kāşgarî has <i>kur</i> “kuşak, kemer”, a belt or girdle, and the verb <i>kurlamak</i> , “kuşak yapmak ve bağlamak”, to make a belt and to bind, together with <i>kurluk</i> and <i>kurlanmak</i> . 狐鹿姑 captured the Han general Li Guangli and held the confederation together through a hard decade, so an epithet of binding fits what the record says he did. Compare the Göktürk ruler Kapgan Kagan, <i>kapgan</i> “the encloser”. Two objections in an earlier draft of this entry were wrong and are withdrawn. It described the <i>-r</i> as a segment Later Han cannot write and the <i>-k</i> as a discarded coda, and said the two pulled in opposite directions. Neither holds: leaving a liquid unwritten before an open character is the majority spelling at 54%, and a written coda reaching forward to the next onset is the commonest use of a written coda at 37%. Both were judged by ear before they were counted. What is not settled, and it is the shape of the word rather than any of its sounds. <i>Kurluga</i> is not an attested headword, and neither is <i>kurluka</i> ; the frame returns no word at all. More than that, the shape is not one Turkic builds: the suffix <i>-IXk</i> / <i>-IXg</i> is word-final, 380 headwords end in it, while only one headword in 9,343 ends in <i>-luka</i> and none in <i>-luga</i> . The well-formed Turkic word in this frame is <i>kurluk</i> or <i>kurlug</i> “girded”, which 狐鹿 write between them with 鹿's own <i>-k</i> as the final consonant, and which leaves 姑 unaccounted. That is the same structural question as at 虛閭權渠: whether a longer Xiongnu name is one word or a name followed by a title element. It is not answered here, and this reading assumes the first.				
壹衍鞮	ga jan ^B te [†]	ga yan te	Kayantı	avalanche
Huyandi				
PHONETICS 壺 <i>ga</i> → <i>ka</i> : given a Later Han voiced g-, the source sound is more often k than g, so the devoicing is normal rather than a liberty. 衍 <i>jan</i> → <i>yan</i> : direct, coda written. 鞮 <i>te</i> → <i>tr</i> : direct. All three steps are default or better, phonetically one of the cleaner readings here.				
REASON <i>kay-</i> “to slide, to slip away”. Parallel in sense to Bayezid I's <i>Yıldırım</i> : a natural force taken as a regnal epithet.				
虛閭權渠	hĩa lia gyan ɣĩa	kĩa lia gyan ɣĩa	Kalkan	shield
Xulüquanqu				

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 虛 <i>hia</i> → <i>ka</i>: the second of the character's two Later Han readings, which §8.5 selects. <i>Kalkan</i> has back vowels throughout, so its velars are <i>q</i>, and Table 10 measures a Turkic back <i>q</i> written with an <i>h</i>- character 43 times in 66, 65.2%, against 20 in 66 for an aspirated one. On the aspirated reading the first three characters cost 1 in 150; on this one they cost 1 in 6. 閭 <i>lia</i> → a bare <i>-l</i>, the onset direct at 97 of 100 but the vowel writing nothing, which is the final-character device measured at 45.7%. 權 <i>gyan</i> → <i>kan</i>, with the <i>-n</i> on the page. 渠 is left unaccounted, and the whole reading costs 1 in 1,138. That character is a real objection rather than a footnote. It writes a full syllable elsewhere in this record, the <i>ka</i> of the reading proposed at 且渠, with its velar onset and open vowel doing work; and it recurs as the second element of Xiongnu compounds across the histories, 且渠 as an office, 顓渠 in the consort title 顓渠閼氏, and 雕渠 in the personal name 雕渠難. A character that behaves that way in four compounds is not idle in the fifth by nature. Nor does the corpus license leaving it idle: of the 152 pairs in 1,110 that use more characters than syllables, 77 are a Chinese semantic suffix appended to a finished transcription, 鬼 “demon”, 鳥 “bird”, 國 “country”, 城 “city”, and 渠 is not a classifier of that kind. No word in the lexicons uses all four characters. The frame returns 0 four-syllable fits, 7 three-syllable fits with 渠 left over, all of them <i>-gAn</i> particples and none of them a ruler's name, and 9 two-syllable fits which need both 閭 and 渠 to be nearly idle. REASON <i>Kalkan</i>, “shield”. The word is in Kāşgarî and in the <i>Codex Cumanicus</i> as <i>Kalkani</i>, “scutum ejus”, and a weapon or armour name is an ordinary steppe formation. What is not settled, and it is more than the usual caveat. The reading survives only if two of the four characters carry almost nothing, and a character contributing a coda consonant alone, as 閭 does here, is measured only as the final-character device and not in this medial position. The likeliest fix is that 虛閭權渠 is two elements rather than one word. 且渠 is an office in its own right in this same record, and 顓渠閼氏 shows the pattern of an X渠 element standing before a title, so 權渠 may be a title following a name in 虛閭 rather than the second half of a single word. That has not been tested, and if it holds this reading is wrong. WE record <i>kalkan</i> as permitted on the sounds and unsupported on the shape.				
握衍胸鞬	ʔɔk jan ^ɐ † guo te [†]	’ok yan guo te	Okyan Kutı	the fortune of the arrow flank
Woyanqudi				
PHONETICS 握 ʔɔk → <i>ok</i>: a source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140, 97.9%, the strongest single correspondence used anywhere in this file, and the <i>-k</i> is written, 47.6%. 衍 <i>jan</i> → <i>yan</i>: a source <i>y</i> written with a Chinese <i>j</i>- is 65 of 106, 61.3%, the vowel is open, 1,127 of 1,634, 69.0%, and the <i>-n</i> is on the page, 82.8%. 胸 <i>guo</i> → <i>ku</i>: this is the weak cell and it is named rather than buried, a source <i>k</i> written with a Chinese <i>g</i>- being 14 of 254, 5.5%; the rounded vowel is 63.9%. 鞬 <i>te</i> → <i>tɪ</i>: a source <i>t</i> written with a Chinese <i>t</i>- is 61 of 241, 25.3%, and the front vowel on a front character is 86.0%. Four characters, four syllables, one each. In order, nothing inserted, nothing discarded, no compression and no device, and every one of the ten cells measured with none assumed. The whole reading costs 1 in 1,252, 16.8% per character, twice the 8.1% per character that the control at 撐犁 gives for a reading the Chinese sources gloss outright. The reading this replaces could not be priced at all, and that is why it is replaced. <i>Kutuġtı</i> asked 胸 to carry two Turkic syllables, <i>kutuġ</i>, where it carries one, and so offered five syllables for four characters. Nothing in the corpora licenses that expansion: the compression device, one character for two syllables, is 93 of 1,110 in the Sanskrit corpus, 8.4%, and it requires the coda of the character to match the place of the following onset, which <i>guo</i> and <i>t</i> do not. The engine returned no rate rather than a price, and a row with no rate is not a proposal. What the shorter form gives up. Only the <i>-uġ</i>-. The reading is otherwise the same word in the same order, and the change is from a form the characters cannot write to one they can. REASON <i>Okyan Kutı</i>, the fortune of the arrow flank. All three elements are Kāşgarî headwords: <i>ok</i> “arrow”, <i>yan</i> “side, flank”, and <i>kut</i> “heavenly favour, fortune”, here with the third person possessive <i>-ı</i>. Ok as a division of the confederation is attested and is what makes a territorial reading available, the <i>On Ok</i>, “ten arrows”, of the Western Turks being the standing example; a wing named by the arrows it fields is the ordinary steppe idiom rather than a construction made for this row. The formation is one this file already uses, which cuts both ways. 孤塗 is read <i>Kutı</i> at 1 in 161, and 撐犁孤塗 is <i>Taŋrı Kutı</i>, “Heaven's fortune”, which the <i>Hanshu</i> glosses outright. <i>Okyan Kutı</i> is that same formation with a different possessor, the flank in place of heaven. That the record demonstrably uses <i>X Kutı</i> is a point in favour of reading it again; that a formation already in the file is an easy target to hit is the objection §6.3 of the paper measures, and it applies here. What is not settled. Whether <i>okyan</i> is one word or two, which the transcription cannot say, since it writes syllables and not word boundaries. Whether the first element is the arrow at all, or the vowel-initial syllable of some other word, 握 carrying only that the syllable began with a vowel and ended in <i>-k</i>. And the price rests on one thin cell, 胸 writing a <i>k</i> at 5.5%, so the reading is no better than that cell: were it to fall the row would fall with it.				
呼韓邪	ha [†] gan ja [†]	ha gan ya	Kaganya	kagan of the flank
Huhanye				
PHONETICS 呼 <i>ha</i> → <i>ka</i>: a source back <i>q</i> written with an <i>h</i>- character, which §8.5 of the paper measures at 43 of 66, 65.2%, against 20 of 66 for an aspirated character. It is the majority spelling, not a substitution needing indulgence. The 27% figure earlier drafts quoted here was the Mongolian rate; the Turkic one is the relevant measurement and it is higher. 韓 <i>gan</i> → <i>gan</i>: exact, coda written. 邪 <i>ja</i> → <i>ya</i>: a source <i>y</i>- takes this character class 57 times in 98, 58.2%, the largest single destination for a source <i>y</i>- though not a majority of the corpus. The written sequence <i>ha gan ya</i> becomes <i>ka-gan-ya</i> on one change. REASON 呼韓邪 commanded the left flank before becoming ruler, so the Chinese record itself supplies the office that the reading names.				
鄯支	tʃit tʃe [†]	çit çe	Çitçi	border-maker, one who carves the boundary
Zhizhi				
PHONETICS 鄯 <i>tʃit</i> → <i>çit</i>: tʃ- is the default affricate vehicle and the <i>-t</i> coda is written. 支 <i>tʃe</i> → <i>çi</i>: default. Both syllables use the majority correspondence and neither requires an inserted or deleted segment. REASON <i>çit</i> / <i>çiz</i>-, of marking or carving a boundary. 鄯支 broke away westward and set up a rival seat, so a name of division fits the career, though it is equally possible that the career suggested the name.				

TABLE S2 The fifteen rulers whose names are not built on the recurring element 若鞬. Reign order follows the *Shiji* and *Hanshu* as reproduced in Appendix A of the paper.

S1.7 The six 若鞮 compounds

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
若鞮 Ruodi	ńak te [†]	nyak te	ínakt	the faithful ones
PHONETICS 若 <i>ńak</i> → <i>ínak</i>: a palatal nasal onset with the -k coda written. 鞮 <i>te</i> → -t: the final stop is not written, and Table 7 of the paper licenses that at 30%. One objection stands and the paper states it: if the source was <i>ínakt</i>, the opening <i>ɪ</i>- is not written and 若鞮 has no room for it. The price depends on which alignment is taken, and the two differ by four orders of magnitude. On the alignment above, with the initial <i>ɪ</i>- simply unwritten, the reading costs 1 in 4,636: the expensive cell is the onset, a Turkic <i>n</i> written with the palatal <i>ń</i>-, and 若 is that character. The automatic pipeline instead puts the <i>i</i> on 若 and the rest on 鞮, which makes 若 write a vowel-initial syllable, 1 of 140, and returns 1 in 218 million. The figure quoted here is the first, because it is the analysis this reading actually proposes; the second is recorded so that the two are not mistaken for a disagreement about the reading rather than about the alignment. The expensive cell is expensive for a reason that can be stated exactly. 若's palatal nasal writes a Turkic <i>c</i> at 11.4% and a Turkic <i>n</i> at 1 of 177, 0.56%, so this reading takes the twenty times more expensive of the two available uses of the character. Priced as a single syllable, 若 is <i>cak</i> at 1 in 26, <i>yak</i> at 1 in 160 and <i>nak</i> at 1 in 536. And the gloss is what pays for it. Of 9,598 headwords in the three lexicons, 559 fit these two characters with both of them working, and their glosses run <i>juldi</i> “washed” at 1 in 94, <i>alti</i> “six” at 1 in 2,021, <i>yigde</i> “oleaster”, <i>yerde</i> “fellow countryman”, <i>yorçı</i> “guide”. Exactly one of the 559 means anything like what the Chinese sources say the word means, and it is this one: the <i>Codex Cumanicus</i> glosses <i>Inak</i> as <i>fidelis</i>, and the <i>Hou Hanshu</i> glosses 若鞮 as 孝. Every cheaper reading of this row is cheaper because it has abandoned the gloss, which is the trade §9.1 of the paper refuses to make at 徑路 and refuses here. REASON One of the two readings argued in the paper itself, at §9, and it is listed here so this supplement can be read as a single list with the standard visible in it. <i>Inak</i> is glossed <i>fidelis</i> “faithful” in the <i>Codex Cumanicus</i>; the <i>Hou Hanshu</i> glosses 若鞮 as 孝 “filial”; the plural in -t is an Old Turkic formation used with titles (<i>tegin</i> → <i>tegit</i>); and Clauson records Chagatay <i>ínak</i> as “a royal representative or senior minister”, so a court title on this root is what the word ordinarily does.				
復株累若鞮 Fuzhulei Ruodi	buk [†] ʈio lui [†] ńak te [†]	buk tio lui nyak te	Büktel ínakt	Büktel, of the ínakt
PHONETICS 復株累 <i>buk ʈio lui</i> → <i>büktel</i>: the b-, the -k coda, the dental and the liquid all fall in place across the three characters, at zero minority steps, with only the front rounded vowel unwritable. One attested word covers the whole prefix rather than an invented two-element compound. 若鞮 → <i>ínakt</i> is treated in §S1.4. REASON Kâşgarî gives <i>büktel</i> as “orta boylu; yassı arkalı, oturamaklı”: of middling height and flat-backed. A name from a bodily description is a well-attested Turkic type; compare the Hungarian royal name <i>Şişman</i>.				
搜諧若鞮 Souxie Ruodi	so [†] gei ńak te [†]	so gei nyak te	Sarg ínakt	the yellow, the fair, of the ínakt
PHONETICS 搜 <i>so</i> → <i>sar</i>: a sibilant onset, which the correspondence counts make obligatory: a source k, q or g was written with a Chinese sibilant in 0 of 2,124 attestations, while Chinese sibilant characters transcribe s, š, č or ś in 98% of 837 cases. The final -r is not written and could not be: Later Han carries no -r in any coda position. 諧 <i>gei</i> → <i>ig</i>: a velar onset, direct. Nothing is inserted, and no character is read as a sound it cannot carry. 若鞮 → <i>ínakt</i> is treated in §S1.4. REASON <i>sarig</i> “sarı, sarı renk” (yellow) is in Kâşgarî, and survives as modern Turkish <i>Sarı</i>, still current both as a name and as an epithet. Color epithets are a documented Turkic naming type: the same dictionary has <i>çakır</i>, which Kâşgarî glosses <i>azraq</i> “blue” and applies to eyes in his own proverb, <i>it çakırı atqa tegir, at çakırı itqa tegmes</i>, “a blue-eyed dog is worth a horse, a blue-eyed horse is not worth a dog”, because such a horse sees badly; the derived verb <i>çakrat</i>- is glossed “to roll the eyes and look blue-eyed”. Modern Turkish <i>çakır</i> is greyish-blue of the eyes, and earlier drafts of this entry glossed it that way; the eleventh-century gloss is the plain colour. <i>Çal</i> “dappled, grey” is proposed for 烏珠留若鞮 on the same footing. Ninety-one attested headwords fit this two-syllable frame, so it does not determine the reading; <i>sarig</i> is offered as the most plausible of them, with nothing in the transcription standing against it.				
車牙若鞮 Cheya Ruodi	tś ^h a [†] ŋa ńak te [†]	ça nga nyak te	Tuŋa ínakt	the tiger, or the hero, of the ínakt

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 車 <i>tʂʰa</i> → <i>to</i>: the first of the character's two tabulated readings, no polyphony argument needed. A source <i>t</i> written with the aspirated <i>tʂʰ</i>- is 67 of 198, 33.8%; the character is open, so any source coda would be unwritten; the vowel is free, a source <i>o</i> written with an open vowel being 103 of 324, 31.8%. 牙 <i>ŋa</i> → <i>-ŋa</i>: a source syllable beginning with a velar nasal takes a Chinese <i>ŋ</i>- character 5 times in 7, and the one intervocalic case is 丁彦多 for <i>tiñanta</i>, where 彦 carries the <i>-ñan</i>-; a source <i>a</i> written with an open vowel is 1127 of 1634, 69.0%. The whole name costs 5.29%, about 1 in 19, against the 1 in 152 that the control at 撐犁 gives for a reading the Chinese sources gloss outright. Step 48 prices every form the two characters permit. The reading this replaces is excluded twice over. <i>Ok</i> required 牙 to write a vowel-initial syllable, which is 0 of 140 in the Sanskrit corpus, 137 of the 140 taking a glottal; and <i>çöge</i> required a source affricate written with the aspirated <i>tʂʰ</i>-, which is 1 of 59, where the vehicle for a source affricate is the plain <i>tʂ</i>- at 37 of 53. It also added a syllable the character does not carry. A likelier form exists and is rejected on sense, not on phonology. On the second reading of 車 the frame admits <i>kaña</i>, from the Codex <i>canga</i>, at about 1 in 3; it is glossed <i>tabulatum</i>, planking, which is not a ruler's name, and it needs the second reading where <i>toña</i> needs only the first. The transcription under-determines here as it does everywhere, and the choice is made on meaning, which §6.3 of the paper measures as the weakest step in the whole procedure. On the spelling. The <i>ŋ</i> of <i>Tuŋa</i> is a single velar nasal, not a nasal followed by a stop, and the <i>-ng</i>- of the modern approximation <i>Tunga</i> renders that one sound. That is why these two characters write the word and 當戶 cannot: a written <i>-ŋ</i> coda followed by a velar onset is the corpus device for a source <i>ŋg</i> or <i>ŋk</i> cluster, 13 cases of 13, and this word has no velar stop in it. REASON <i>Tuŋa</i>, an epithet, in the title slot the other 若鞮 names use. Clauson's headword is <i>toŋa</i>-, with back vowels, and he records that “whether -o- or -u- in the first syllable is uncertain”, so <i>Tuŋa</i> and <i>Toŋa</i> are equally available; this file prints the first, and the transcription cannot separate them either, since 車 carries an open vowel and a source <i>o</i> and a source <i>u</i> fall alike at 31.8%. Two senses, resting on different evidence, and both are recorded here. Kāšgharī glosses the word as a tiger-like beast (<i>Diwān</i> III 368), and glosses the title <i>Toŋa Alp Er</i>, given to Afrasiyab the king of the Turks, as “the heroic man as strong as a tiger”. Outside personal names the eighth- and ninth-century Uyghur texts use it for “hero, outstanding warrior”: the Manichaean <i>éligler xanlar toŋalar</i>, “kings, khans, great warriors” (M III 41), the Buddhist <i>Arcunī toŋa</i>, “the hero Arjuna” (U II 24), and <i>toŋalar begi</i>. Clauson records that the animal sense “is not confirmed by any other authority”, and adds that by Kāšgharī's day the word's meaning had become unknown. That second point rests on an emendation, and the two standard editions of the <i>Diwān</i> disagree. The manuscript reads <i>salla</i>, and Dankoff and Kelly translate it as it stands: “this word has remained with the Turks and its meaning persists among them.” Clauson prints “its meaning became unknown” and marks the change in his own parenthesis, <i>read dalla for salla in MS</i>. Whether the animal sense was live or already fossil in 1072 therefore turns on one letter, and it does not affect this reading either way: on both readings the word was in use as a title, which is the only thing this name needs. The naming pattern is attested, and it is the one this name needs. Kāšgharī says the word is <i>yulaqqab bihi</i>, used as an honorific epithet, gives <i>toŋa xan</i> and <i>toŋa tegin</i>, and glosses the title itself, <i>toŋa alp ar</i>, as “a man, a warrior, as strong as a tiger”. All of this is taken from Dankoff and Kelly's edition rather than at second hand. The shape is the epithet first, the title after it, head last, which is the shape of <i>Alp Arslan</i> and <i>Bilge Kağan</i>. <i>Tuŋa İnakt</i> is that shape, with 若鞮 as the title element the paper argues in §9, and it matches the form the other five 若鞮 rows already take, such as <i>Çal İnakt</i>. Both word orders are Kāšgharī's, and they belong to different registers. The verse he quotes runs <i>alp ar toŋa öldi mü / esiz ajun kaldı mu</i>, “has Afrasiyab died, has the wretched world been left behind”, which is the order familiar in Turkish as <i>Alp Er Tonga</i>. The title is the other way round, <i>toŋa alp ar</i>, and so are <i>toŋa xan</i> and <i>toŋa tegin</i>. Clauson cites the title form and the Turkish tradition remembers the verse form; neither is mistaken. This bears on the reading, because 車牙若鞮 stands in a title slot, and there the <i>toŋa</i>-first order is the attested one. What is not settled. The vowel of the first syllable, which neither Clauson nor the transcription fixes. Whether the epithet is the animal or the quality. Both senses are recorded above, neither is suppressed, and the transcription cannot choose between them. And this word stood at 當戶 in earlier drafts; it is moved here because 當戶 writes a source nasal-plus-velar cluster and this word has a single nasal.				
烏珠留若鞮 Wuzhuliu Ruodi	ʔa tso liu ńak te [†]	’a ço liu nyak te	Çal İnakt	the grey or dappled one, of the inakt
PHONETICS 烏 <i>ʔa</i> is vowel-initial, so the reading begins at 珠 <i>tso</i> → <i>ç</i>, with 留 <i>liu</i> → <i>l</i>: an affricate written with the default <i>tʂ</i>- character and a liquid, both majority correspondences. Nothing is inserted, and no character is read as a consonant it does not carry. REASON <i>çal</i> is “alaca, kır” in Kāšgharī: dappled, grey. Color epithets are a persistent Turkic naming pattern, as <i>çakır</i> shows in the same dictionary: Kāšgharī glosses it <i>azraq</i> “blue” and applies it to eyes in a proverb about a blue-eyed dog and a blue-eyed horse. He also records <i>Çaqır Toŋa Xan</i> as a ruler's name, so the colour word is attested both as a gloss and in use as a name element.				
烏累若鞮 Wulei Ruodi	ʔa lui [†] ńak te [†]	’a lui nyak te	Uluğ İnakt	the Great, of the İnakt
PHONETICS 烏 <i>ʔa</i> → <i>u</i>:- a source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140, 97.9%, which is the strongest single correspondence used anywhere in this file. 累 <i>lui</i> → <i>-lu</i>: a source <i>l</i> is written with a Chinese <i>l</i>- 97 times in 100, 97.0%, and the vowel is rounded on both sides, 207 of 324, 63.9%. The final <i>-ğ</i> is not written, which the coda table of §8 puts at 9 of 23, 39.1%, the ordinary treatment of a source velar coda. Two characters, two syllables, nothing inserted and nothing discarded. The element costs 23.7%, about 1 in 4. That figure is not comparable with a whole-name figure such as the 1 in 4 for <i>Köngen</i> or the 1 in 152 of the control at 撐犁: it prices a two-syllable element, and a short string scores high by construction, since every further character multiplies in another rate below 1. 若鞮 is not priced again here, being the one title element §9 argues once for all six rows that carry it. REASON <i>Uluğ</i> “great”, from the Codex Cumanicus <i>Oulu</i>, glossed <i>magnus</i>. The word is among the most common elements in Turkic titlature, which is both the strength of this reading and its weakness: a very common word is an easy target to hit, and §6.3 of the paper measures choice by meaning as the weakest step in the whole procedure. The same word was read at 呼都而尸道皋若鞮 in earlier drafts and is withdrawn there. Step 54 measured what separates the two rows. This one has a glottal character for the vowel-initial syllable and a liquid character for the <i>-l</i>-; the long name has neither, and no character in it carries a liquid at all. Where one word appears twice in a record, the transcription and not the sense has to say which occurrence is real, and here it does. What is not settled. Whether the word is a title, an epithet or part of a personal name. The record gives no way to tell, and the other five 若鞮 rows are equally silent on it.				
呼都而尸道皋若鞮 Huduershī Daogao Ruodi	hɑ [†] tɑ ńə [†] si dou ^{ɞ†} kou ńak te [†]	hɑ tɑ nyɪ si dou kou nyak te	Kartay Bedük İnakt	the great elder, with the title

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 呼 <i>ha</i> → <i>kar</i>: a Turkic back <i>q</i> written with an <i>h</i>- character is 43 of 66, 65.2%, the rate §8.5 of the paper measures on Ligeti's glossaries because Sanskrit has no <i>q</i> at all and the Later Han corpus cannot price the choice; <i>kartay</i> has back vowels throughout, so its initial is a <i>q</i>. The vowel is open, 1,127 of 1,634, 69.0%, and the <i>-r</i> is not written, which a source liquid coda is in 21 of 37, 56.8%, Later Han carrying no <i>-r</i> in any coda position. 都 <i>ta</i> → <i>tay</i>: a source <i>t</i> written with a Chinese <i>t</i>- is 61 of 241, 25.3%, and the vowel is open again, 69.0%. 而 <i>riə</i> → <i>-y</i>: the character carries the final <i>-y</i> of <i>kartay</i> through the final-character device, which the corpus puts at 45.7%, and a source <i>y</i> written with a palatal nasal is 2 of 106, 1.9%. 尸 <i>ši</i> → <i>be</i>: a source <i>b</i> written with a Chinese <i>ś</i>- is 6 of 129, 4.7%, and the vowel is front on a front character, 86.0%. 道 <i>dou</i> → <i>dük</i>: a source <i>d</i> written with a Chinese <i>d</i>- is 135 of 183, 73.8%, and the rounded vowel is 96.9% on the Ligeti control that Table 9 uses for the front rounded series. 皋 <i>kou</i> → <i>-k</i>: the character carries the final <i>-k</i> of <i>bedük</i> through the same device, and a source <i>k</i> written with a Chinese <i>k</i>- is 208 of 254, 81.9%. Every one of the six characters carries sound. Nothing is inserted, nothing is discarded, the order is the order on the page, and no character is read as a sound it cannot carry. The whole reading costs 1 in 243,285, 12.7% per character. That figure cannot be set beside the control's 1 in 152 directly, because a long string is expensive by construction and every further character multiplies in another rate below 1; the comparable quantity is the per-character one, where the control at 撐犁 stands at 8.1%. Six characters at the control's rate would cost about 1 in 3.5 million, and this reading is some fourteen times cheaper than that. 若鞮 is not priced again here, being the one title element §9 argues once for all six rows that carry it. The two weak cells are named rather than buried. They are 而 at 1.9% and 尸 at 4.7%, and between them they cost more than the other four characters together. 而 is expensive whatever is done with it, and that is the finding this row rests on. The character carries a palatal nasal onset in both its tabulated readings, <i>riə</i> and <i>rii</i>, which differ only in the vowel, so no polyphony argument reaches it. Turkic has no palatal nasal at all: what Turkish spells <i>ñ</i> or <i>ng</i> is a velar nasal and answers to Sanskrit <i>ṇ</i>, not to <i>ñ</i>. Only four Turkic sounds reach this character, and these are all of them: <i>c</i> at 4 of 35, 11.4%, <i>j</i> at the same rate, <i>y</i> at 2 of 106, 1.9%, and <i>n</i> at 1 of 177, 0.56%. Old Turkic has no <i>j</i>. Against that, the corpus licenses a character that writes nothing at 152 of 1,110, 13.7%. Every sound 而 can carry costs more than discarding it, which is why earlier searches on this name kept returning readings that left it idle, and it is the reason the string resisted for so long. This reading pays the 1.9% rather than the 13.7% and buys a character for the money. It is one of only two characters in the whole record with that onset, and this name contains both. Of the 81 distinct characters the record uses, 而 and 若 alone carry a palatal nasal, and 若 is the title element closing this same name, where §9 reads it as a Turkic <i>n</i> on the palatal <i>ri</i>- and names that as the expensive cell of the reading. The hardest name in the record is the one that has to spend both. The reading earlier drafts gave this name, Öz-Uluğ İnakt, stays withdrawn, and not because something cheaper turned up. It failed three measurements and still does. A source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140; it takes a velar 0 times in 140 and an <i>h</i>- 0 times in 140, and both its elements sat on velar or <i>h</i>- characters, 呼 <i>ha</i> and 皋 <i>kou</i>. A source <i>l</i>- is written with a Chinese <i>l</i>- 97 times in 100, and a source liquid onset of either kind 262 times in 288, while not one character in this name carries a liquid onset, so the <i>-l</i>- of <i>uluğ</i> would have gone unwritten in the one position the corpus almost always writes it. And it left four of the eight characters carrying nothing, where across 1,110 pairs in the Sanskrit corpus the count of transcriptions with four or more spare characters is 0. <i>Uluğ</i> stays at 烏累若鞮, where the transcription supports it. What a vowel-initial first element would still cost, since that is the shape the withdrawn reading wanted. 呼 carries an <i>h</i>-, so a vowel-initial word could in principle be written by putting a prosthetic <i>h</i>- in front of it, and <i>hoz</i> prices at 1 in 5, the cheapest single syllable this character offers. But Kâşğarî has exactly five headwords beginning with <i>h</i>- and they are a closed class: <i>hana</i> and <i>hata</i>, the address forms of “mother” and “father”, listed together at I 32 and each cross-referred to its <i>h</i>-less form, then <i>heç heç</i>, a call for urging horses, <i>hoç hoç</i>, a call used in herding goats, and <i>hukubarı</i>, a potter's clay. An initial <i>h</i>- in his dictionary belongs to vocatives and to animal calls and to nothing else, and Clauson describes the prosthetic <i>h</i>- of later Türki as recently acquired. So <i>höz</i> cannot be built. <i>Kartay</i> needs no such device, its initial being a consonant the character writes directly. The name divides where the Hanshu divides it. 若鞮 is the title the same sentence glosses as 孝, and three lines after giving the full form the <i>Hanshu</i> calls the same ruler 呼都而尸單于, dropping 道卑若鞮 altogether, so the record itself treats 呼都而尸 as enough to identify him. 尸 is an Eastern Han element the <i>Shiji</i> does not use once; in the <i>Hou Hanshu</i> it appears thirteen times and every one is in the fixed sequence 尸逐, the standing element of a dozen Southern Xiongnu regnal titles. On that distribution this reading takes 尸 as a syllable of the name rather than as that element, which is a claim the row makes and cannot prove, since 尸道 occurs nowhere else. REASON <i>Kartay</i> <i>Bedük</i>, the great elder. <i>Kartay</i> is an attested word, and Clauson has it. Under his headword <i>1 karı</i>: “old” he cites the Kipchak <i>Kitäb al-Tuhfa</i> (20b.3), where for someone older than oneself the words given are <i>abıřka</i> and <i>kartay</i>. So <i>kartay</i> is an elder, a senior, one older than oneself. Two different words are spelled <i>kart</i>, and this reading rests on the second. Clauson separates them. The first is <i>kart</i> “the scab on a wound, an ulcer”, Arabic <i>al-qarḥ</i>, which is Kâşğarî I 342 and the one the <i>Dıwân</i> index glosses <i>yara</i>; from it come <i>kart er</i>, an ill-tempered man, the verbs <i>karta</i>-, <i>kartal</i>- and <i>kartan</i>-, and probably <i>kartal</i>, which Clauson suggests was first “a spotted or striped bird”. The second is <i>kart</i> “old”, of which he writes that it “first appeared in Western dialects in the medieval period and is still current in NC, NW, SW; it is clearly cognate to <i>1 karı</i>: but cannot morphologically be derived from it”: Xwarazmian XIII <i>bir kart kiři</i>, an old man, Kipchak XIV <i>kart</i>, old. <i>Kartay</i> belongs to the second. On the date of the Kipchak witness. Clauson labels his <i>Tuhfa</i> citations xv. The date is disputed: Al-Turk's survey of the manuscript, Beyazıt Umumi Kütüphanesi, Veliyüddin Efendi 3092, records that opinions differ, and reports Kuun's comparison of its hand with the autograph of as-Safadi, a pupil of Abü Ḥayyân born at Safad late in the thirteenth century, which would place it in the fourteenth. Nothing in this row turns on it, since the Khwarezmian attestation of about 1310 is earlier than either date and is not in dispute. The word is not a hapax, and it is not the <i>-tay</i> of <i>turumtay</i>. An earlier draft of this entry compared it with Kâşğarî's <i>boxtay</i> and <i>turumtay</i>, where Clauson identifies a rare noun-forming suffix <i>-ta:y</i> “which has no relation to the Mong. P.N./A. suffix <i>-tay/-tey</i>”. That comparison is withdrawn, and it is withdrawn without a replacement, because no authority consulted here analyses <i>kartay</i> at all. Clauson cites it twice and parses it nowhere; Erdal's grammar records neither suffix for Old Turkic; Gülensoy, as Cin reports him, derives <i>kart</i> and stops there. The simpler reading of the two available is that <i>kartay</i> is <i>kart</i> with the denominal verb suffix that gives Kazakh <i>ulğay</i>- “to advance in years” and Bashkir <i>olagayıv</i> beside <i>kartayıv</i>, the noun being that stem in its bare form, which is how the Kipchak lexicographers cite verbs. The other reading, a noun in the <i>-ta:y</i> of <i>boxtay</i>, is not impossible and is simply not evidenced. Either way the morphology of this row rests on the stem and not on a name-forming element, which is less than the earlier draft claimed. What the word does have is depth and spread. Cin's survey of the vocabulary of old age from the Orkhon inscriptions onward gives the Kipchak <i>Tuhfa</i> three forms from the same stem, <i>qart</i> “yaşlı”, <i>qartın</i> “yaşlı” and <i>qartay</i> glossed <i>kart</i>, <i>ihtiyar</i>, old, aged, which is the noun this reading needs and which agrees with Clauson's independent reading of the same manuscript. And it is a century and a half older than that. The Khwarezmian <i>Qisasü'l-Enbiyâ</i> of about 1310 has it in running text rather than in a glossary: <i>İncilni ÖArab tilinçe bitimiř erdi kartayıp erdi</i>, “he had written the Gospel in Arabic, he had grown old” (KE 194r/5). The spread is the strongest part. The form is in Kazakh <i>kartay</i>-, Kyrgyz <i>kartayu</i>, Bashkir <i>kartayıv</i> and <i>kartaygan</i>, Tatar <i>kartaygan</i>, Karay <i>kartay</i>-, and in Anatolian dialect, where the Derleme Sözlüğü records <i>kartay</i> “yaşlanmak” for İstanbul and <i>kartı</i>- for Afyon. A form that reaches from Kazakh to İstanbul is not plausibly a single coinage of fifteenth-century Mamluk Egypt. The root is older still, and it is in the inscriptions. Clauson's <i>karı</i>:- “to be, or become, old” is Türkü VIII, and Cin cites the same layer, <i>Säkiz on yaşayur qarı erti</i> “he was an old man of eighty” and <i>Karıyuk biz</i> “we have grown old”, beside Kâşğarî's <i>er karıdı</i> and his proverb <i>arslan karısa</i>, “when a lion grows old”. <i>Bedük</i> is Kâşğarî's <i>büyük</i>, great (I 93, 360, 385, 499, 500), the ordinary word for it, and “the great” as the second element of a ruler's epithet needs no further argument. The shape is the epithet first and the title last, which is the shape of <i>Alp Arslan</i> and of the other 若鞮 rows in this file, <i>Çal İnakt</i> and <i>Tuğa İnakt</i>. The objection to this reading is chronological rather than phonetic, and it is a real one. Clauson dates <i>kart</i> “old” to the medieval West. This name was written between 18 and 46 CE. Proposing a Kipchak form of the fourteenth and fifteenth centuries for a first-century transcription reaches across thirteen centuries and into the wrong dialect area, and the price cannot see that, because the engine scores letters and not dates. The objection reduces to one disputed morpheme, and two authorities disagree about it.				

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
Everything in the word except the <i>-t</i> is Old Turkic: the root <i>kari-</i> is Türkü VIII, in the inscriptions themselves, where the line Clauson quotes pairs the two ideas this name pairs, <i>özüm kari: boltım uluğ boltım</i>, “I myself have become old, I have become great”. Clauson holds <i>kart</i> to be “clearly cognate to <i>1 kari:</i>” but says it “cannot morphologically be derived from it” and dates it to the medieval West. Gülensoy derives it directly and places it earlier. His etymological dictionary gives <i>kart</i> as <i>kari-</i> “yaşlanmak” plus <i>-t</i>, and Cin cites him for it under Uyghur Turkic, alongside the Uyghur <i>ķurta</i>, <i>ķurtğa</i> and <i>ķartğa</i>, “old, aged, long-lived”. If Gülensoy is right the stem is Old Uyghur and the chronological objection to this row largely dissolves; if Clauson is right it stands. This entry does not adjudicate between them. It records that the reading needs Gülensoy’s derivation to be correct, names the disagreement, and leaves it where a reader can weigh it. What can be said without taking sides is that the objection has narrowed from the whole word to a single suffix. The old form cannot be written by these characters and the late one can. <i>Kari</i> is blocked outright: 都 would have to write <i>ri</i>, and a Chinese dental stop character writing a source liquid onset is a measured zero of the inventory-gap kind, the same zero that retires readings elsewhere in this file. So the transcription admits the medieval shape and excludes the ancient one, which is an awkward result and is set down here rather than left for a reader to find. What else the characters permit, so that the choice can be weighed rather than taken on trust. Scoring all 5,656 single-word headwords of Kāšgarī and the <i>İrk Bitig</i> against every contiguous group of these six characters and recombining, the readings in which every character carries sound are few and lie close together: <i>Kaş Tay Bedük</i> and <i>Kas Tay Bedük</i> at 1 in 178,169, 13.3% per character, <i>Ka Tay Bedük</i> at 1 in 138,081, 13.9%, <i>Ka Tay şiltük</i> at 1 in 90,691, 14.9%, and this reading at 1 in 243,285, 12.7%. The word boundary is free to the engine, which scores letters, so <i>Kastay</i> and <i>Kaştay</i> price exactly as <i>Kaş Tay</i> does; neither is a headword in any of the lexicons, and neither is offered. The choice among the survivors is therefore lexical and not phonetic. <i>Kartay Bedük</i> is preferred because both its elements are attested words and the pair reads as a ruler’s epithet, where a jade colt or a horn inkwell does not. Cheaper readings exist and are refused for a stated reason. If 𐰽 may be discarded at 13.7%, <i>Hata şī Dük</i> costs 1 in 748, 33.2% per character, four times cheaper than the control and the cheapest thing these characters admit. It is not offered: <i>dük</i> is Kāšgarī’s “that much, a few” and is not a name, and the reading leaves idle the very character this row exists to account for. A reading that buys 𐰽 is worth more than a reading that is cheap without it. What is not settled. Whether <i>kart</i> “old” existed in the first century, on which Clauson’s dating tells against this reading and his etymology tells for it, and nothing here decides it. Whether 尸 is a syllable of the name or the 尸逐 element of the Southern Xiongnu titles, which the distribution makes likely and 尸道 leaves unproven. Whether the epithet is a personal name, a title, or both, on which the record is silent for all six 若鞮 rows alike. And whether any of it is Turkic, on which this entry, like the rest of the file, is a proposal and not a finding.				

TABLE S3 The recurring element itself, then the six later chanyu whose formal names end in it. Only the shared element is argued for in the paper; the material preceding it in each name is a proposal like any other in this supplement.

s1.8 The ruling clan name, in its two written forms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
虛連題 Xulianti	hia lian ^h de ⁺	ha lian de	Karındaş	kinsmen, those of one womb
PHONETICS 虛 <i>hia</i> → <i>ka</i>: the second of the character’s two Later Han readings, and §8.5 of the paper measures why that one. Turkic distinguishes a front <i>k</i> from a back <i>q</i> and Sanskrit has neither the distinction nor a <i>q</i>, so the Later Han corpus cannot price the choice and Table 10 prices it on Ligeti’s glossaries instead: a Turkic back <i>q</i> is written with an <i>h</i>- character 43 times in 66, 65.2%, while a front <i>k</i> takes an aspirated character 31 times in 35. <i>Karındaş</i> has back vowels throughout, so its initial is a <i>q</i>. On the aspirated reading <i>k^hia</i> the onset cell falls to 19 of 254, 7.5%, and the whole name to 1 in 211, so this row depends on Table 10 and says so. The vowel is open, 1,127 of 1,634, 69.0%. 連 <i>lian^h</i> → <i>rin</i>: a source <i>r</i> written with a Chinese <i>l</i>- is 185 of 208, 88.9%, the vowel is front, 349 of 406, 86.0%, and the <i>-n</i> is on the page, which a source <i>-n</i> is in 94 of 111, 84.7%. 題 <i>de</i> → <i>daş</i>: a source <i>d</i> written with a Chinese <i>d</i>- is 135 of 183, 73.8%, and the <i>-ş</i> is unwritten, which a source sibilant coda is in 31 of 40, 77.5%, Later Han having no sibilant coda to write it with. The weak cell is named rather than buried. It is the vowel of the third syllable: an <i>a</i> written with a front-vowel character is 405 of 1,634, 24.8%, and that one cell costs more than the rest of the name together. The whole reading costs 1 in 24, against the 1 in 152 that the control at 掣犁 gives for a reading the Chinese sources gloss outright. Three characters, three syllables, in order, with nothing inserted, nothing discarded and no metathesis. Only two words in the lexicons fit the frame at all, and the other one, <i>kirindi</i> “crumbs”, prices at 1 in 56 and is not a name. REASON Karındaş, “those of one womb”, kinsmen. The word is <i>karn</i> “womb” with the suffix <i>-daş</i>, and Kāšgarī glosses it <i>kardeş</i>; it is the ordinary modern Turkish word for a sibling. A ruling lineage naming itself by kinship is an ordinary formation and needs no epithet argument, which is the weakness §§1.3 names as this supplement’s main one. Why the longer form is read and the shorter one is not. The two-character form 掣犁 begins with a liquid and no Turkic word does, so the shorter record cannot be read as a word at all; the three-character form has the liquid medially, where the constraint disappears. That entry sets out the counts and the chronology. What is not settled. The <i>q</i> rate is measured on Ligeti’s Ming glossary rather than on Han material, because Sanskrit has no <i>q</i> at all; §8.5 states this and uses it as a cross-period bound in the same way Table 9 uses the front rounded vowel. The vowel of the third syllable is the costly cell and carries most of the price. 虛 may be a separate element standing before a two-syllable clan name rather than the name’s first syllable, in which case the core is the unreadable part and this reading is wrong; the character is used transcriptionally in only two other places in the record, 虛閭權渠 and the personal name 虛渠, and in both it stands at the head of a name, which is where a syllable of the name would stand and where a title would stand equally well. WE offer this as permitted rather than confirmed.				
掣犁 Luandi	lyan te ⁺	lyan te	-rındaş	the last two syllables of Karındaş, the first lost

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 攀 <i>lyan</i> and 鞮 <i>te</i> are the same two syllables as 連題 in the longer form of this name: 攀 and 連 are both an <i>l</i> onset with <i>ian</i>, and 鞮 and 題 differ only in voicing and tone. No Turkic word fits the frame. A liquid onset, a written nasal coda and a following dental returns zero of the 9,343 headwords in the three lexicons, and the reason is structural rather than accidental: Old Turkic does not begin words with a liquid. Only 64 of those 9,343 headwords start with <i>l</i>- or <i>r</i>-, 0.7%, and every one of them is a loanword. A vowel-initial first element is excluded outright. A Chinese <i>l</i>- character writing a source syllable that begins with a vowel is 0 of 140 in the Later Han corpus; 137 of those 140 took a glottal instead. What a Chinese <i>l</i>- does write is a source <i>r</i>, 185 of 208, 88.9%, or a source <i>l</i>, 97 of 100, 97.0%. So the syllable behind 攀 began with a liquid, and a Turkic word cannot. REASON Not a word on its own, but the last two syllables of one. On the reading proposed at 虛連題 this is <i>-rindaş</i>, with 攀 taking <i>rin</i> and 鞮 taking <i>daş</i>, and the first syllable absent. Priced as two characters and two syllables it costs 1 in 140. That is what is offered for this row, and it is offered as a fragment rather than as a word. The proposal is not that the Xiongnu clan was called <i>rindaş</i>, which is not a Turkic word and could not be one; it is that these two characters write the last two syllables of the word the three-character form writes in full, and the price is the price of that claim. The unreadability is the argument. A genuine two-syllable Turkic name could not have this shape, because the shape begins with a liquid and Turkic words do not. Something is missing from the front, and the three-character form supplies exactly what is missing. The chronology does not run against this, though at first sight it appears to. The <i>Shiji</i> does not give the clan name at all: its account of Xiongnu custom says 有名不諱而無姓字, that they have personal names, not tabooed, and no surname or courtesy name, and 攀鞮 enters the <i>Shiji</i> only through a later commentary quoting the <i>Hanshu</i>. The two forms on record are therefore the <i>Hanshu</i>'s and the <i>Hou Hanshu</i>'s, and the <i>Hou Hanshu</i> chapter is the account of the Southern Xiongnu, whose court the Eastern Han dealt with directly. That the later chapter is not copying the earlier one can be shown in the same passage. Where the <i>Hanshu</i> writes the sky-and-son title 撐犁孤塗, the <i>Hou Hanshu</i> writes 撐犁孤屠, three of the four characters different for the same two words, and Li Xian's commentary quotes the <i>Hanshu</i> forms and closes 與此不同, "this differs from the present text". Where the <i>Hanshu</i> names three noble clans, the <i>Hou Hanshu</i> names four, adding 丘林氏, which the same commentary also flags. The later notice is a second transcription taken by different officials in direct contact, not a corrupted copy of the first, so a fuller form in the later text needs no special pleading. What is not settled. That the two forms are one name is the commentator's identification and not an independent finding; if they are two different names, the argument above does not apply and this row simply has no reading.				

TABLE S4 攀鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*. Both are taken here as the same name, which is the usual view.

S1.9 Titles and lexical items

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
單于 chanyu	džan [†] wa	can va	Çarvuş	army commander; the earlier form of çavuş
PHONETICS Two characters, 單 with a written coda and 于 open. 單 → <i>çar</i>- takes the second of the character's two Later Han readings, the dental <i>tan</i>, for the onset: a source voiceless affricate written with the voiced <i>dž</i>- of the first reading is 0 of 53 in the Sanskrit corpus, while with a dental onset it is 8 of 48, 16.7%. For the coda it takes the Old Chinese value, Baxter and Sagart's <i>*dar</i>, whose liquid becomes the <i>-n</i> of the later table; step 43 measures how often the two reconstructions differ in this way across the record. 于 <i>wa</i> → <i>-vuş</i> is the best value this character offers: a source <i>v</i> written with a Chinese <i>w</i>- character is 9 of 65, 13.8%, against 1 of 55 for a source <i>b</i> and 0 of 140 for a vowel-initial syllable. The final <i>-ş</i> is unwritten, which a sibilant coda is in 27 of 36 cases, 75%, since Later Han has no sibilant coda. The liquid is what makes the form possible. Across the 8,930 headwords of the Kâşğarî index and the Codex, <i>-rv-</i> and <i>-rb-</i> occur 18 times in native words (<i>burba</i>-, <i>karva</i>-, <i>kurbaka</i>, <i>turbi</i>), while <i>-nv-</i> occurs 0 times and <i>-nb-</i> 5 times, all in loanwords: a liquid before a labial is Turkic, a nasal before one is not. Two shapes the frame excludes outright, whatever word is tried. A velar first element is not available: a voiced affricate writing a velar stop appears once in 220 in the Sanskrit corpus and that one case is a translation mislabelled as a transcription, and characters that write <i>kan</i> exactly, 干 <i>kan</i> and 汗 <i>gan</i>^c, were available and not used. A vowel-initial second element is not available either: a glide character writing a vowel-initial source syllable is 0 of 140. REASON <i>Çarvuş</i>, the earlier form of <i>çavuş</i>, "army commander". A native Turkic title of the first rank. Clauson does not mark it foreign, unlike <i>beg</i>, <i>xan</i> and <i>xağan</i>, and Doerfer 1055 has it travelling outward, to Persian and thence to Urdu and Lezgian. It is attested in the eighth-century inscriptions in the highest possible company: <i>bilgesi çavuşı erti</i>, "he was his Counsellor and Army Commander" (Ix. 17), and at Tonyukuk 7 the speaker is Tonyukuk himself. Clauson glosses it "originally the principal military officer of a xağan, the military counterpart of <i>bilge</i>", and adds that "as time went on the post gradually lost status", ending as the modern Turkish sergeant; Kâşğarî defines the office as the one who orders the ranks in battle and restrains the troops from outrages (I 368), and his form of the word is <i>çaβuş</i>, with a bilabial fricative. Why it is proposed here. Founders of dynasties commonly bear military titles: Osman Gazi, Alp Arslan, Alp Tegin. And a supreme title that began as a military one is an ordinary trajectory, as with Latin <i>imperator</i> "commander" and <i>dux</i>. The Türk inscriptions are a thousand years later than the Xiongnu, so the decline Clauson describes was already under way when they were cut, and at the earlier stage the office Clauson calls the principal military officer of a xağan could have been the ruler's own title rather than his officer's. What is not settled. <i>Çarvuş</i> itself is not recorded in the later Turkic sources. Clauson's <i>Etymological Dictionary</i>, Erdal's <i>Grammar of Old Turkic</i>, Kâşğarî's <i>Diwân</i>, the <i>Kutadgu Bilig</i>, Doerfer's <i>TMEN</i> and Tekin's edition of the <i>Irğ Bitig</i> all carry <i>çavuş</i> or <i>çaβuş</i> and none carries a form with the liquid. On this reading the <i>-r-</i> was lost before the eighth century, which is a thousand years after the name was written down.				
撐犁孤塗單于 chengli gutu chanyu	daŋ lei [†] kua da džan [†] wa	dang lei kua da can va	Taŋrı Kutı Çarvuş	Heaven-Fortune Chanyu, the full ceremonial title

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS Six characters forming the full ceremonial title, and the only item in the record the Chinese sources gloss in full. It decomposes into three units that each have their own row here: 撐犁 <i>ḍaŋ lei</i>, 孤塗 <i>kua da</i> and 單于 <i>dʒan wa</i>. The reading proposed for the compound is simply the three parts in order, <i>Taŋri Kuti Çarvuş</i>, and it is entered here so the file has no blank rows: whatever the three parts are read as, the whole follows, and it carries no argument the three rows do not. Priced across all six characters it costs 1 in 19,422,293, which is what a six-character string costs when every one of its cells is ordinary; the per-character figure, 8.0%, is the one to read, and it is the same as the control's. The title is transcribed twice, and not from the same hearing. The <i>Hanshu</i> writes 撐犁孤塗 and the <i>Hou Hanshu</i> writes 控犁孤屠, three of the four characters different for the same two words, each with its own gloss in its own text; Li Xian's commentary in the <i>Hou Hanshu</i> quotes the <i>Hanshu</i> forms and closes 與此不同, “this differs from the present text”. Two independent Chinese renderings of one foreign phrase, agreeing on the sounds and disagreeing on the characters, is the strongest kind of evidence this record offers, and it is what makes this item the calibration point for the whole file. The characters are in Schuessler's table under shapes the record does not use. 撐 is present and 撑 and 控 are not; 犁 is present and 犁 is not. Earlier work here recorded 撐 as absent in both shapes, which was wrong, and the row was left unpriced on that mistake. Folding the variants gives all four characters a reading and the row prices. REASON The <i>Hanshu</i> glosses the phrase as “Son of Heaven, the Chanyu”, with 撐犁 = 天 “heaven” and 孤塗 = 子 “son”. That makes it the single most valuable item in the record, because both the form and the meaning are recorded, and it is the evidence on which most of the language-attribution argument has rested for a century. What it is worth as a control. Because the meaning of the first element is stated rather than inferred, the price of its correct reading measures the machinery rather than the reading: <i>tangri</i> at 撐犁 costs 1 in 152, 8.1% per character. That is the number against which the other whole-name prices in this file should be read. It is not the same quantity as the single-cell rates §8.1 of the paper uses, such as the 1 of 37 for a source <i>-r</i> against a written <i>-n</i>: a cell rate is one measurement and a name price is the product of six or nine of them, so a name price falls with length and cannot be compared with a cell. Earlier drafts of several entries in this file made that comparison, and it has been corrected throughout.				
撐犁 chengli	ḍaŋ lei [†]	dang lei	Taŋri	heaven, sky, god
PHONETICS 撐 <i>ḍaŋ</i> → <i>tang</i>: a retroflex <i>ḍ</i> for a source <i>t</i> is routine and the <i>-ŋ</i> coda is written, 18 of 24. 犁 <i>lei</i> → <i>ri</i>: a source <i>r</i> written with a Chinese <i>l</i>- is 185 of 208, 88.9%, with the vowel front, 349 of 406, 86.0%. The spelling picks one of the two reconstructed variants, and it is the back one. The word is reconstructed in two ablaut forms, front <i>tengri</i> and back <i>tangri</i>. Both are continued: the front one by Old Turkic, where the derivatives settle it by harmony, <i>tengriken</i> “devout” taking <i>-ken</i> and <i>tengridem</i> “god-like” taking <i>-dem</i>, and by Kāšgārī; the back one by Oghuz and Chuvash, giving Turkish <i>Tanrı</i>, Chuvash <i>Tură</i> and Bulgarian <i>Тангра</i>. 撐 is an open-vowel character, and in the Later Han corpus an open vowel writes a source <i>a</i> 1,127 times in 1,634, 69.0%, against a source <i>i</i> or <i>e</i> 27 times in 406, 6.7%. Priced on the two variants, <i>tangri</i> costs 1 in 152 and <i>tengri</i> costs 1 in 1,579, a factor of ten, and the whole of that factor is the one vowel cell. The scribe was not short of the other option. Schuessler's table holds 154 characters with the nucleus <i>eŋ</i>, 72 with <i>ieŋ</i> and 41 with <i>eŋ</i>, so a front vowel before a velar nasal was available in quantity and was not used. This is not a new result and is not offered as one. The standard reconstruction already cites this very transcription as evidence for the back variant, so what happens here is that the channel measured in this paper, applied to the characters alone, returns the answer the comparative method reached from the daughter languages. That agreement is the point of the row. The proposal is given in the phonetic spelling, Taŋri, and the reason is a measurement rather than a convention. 撐 carries a velar nasal in its own coda. Writing a velar <i>ŋ</i> with it is 72.7%; writing a dental <i>n</i> with it is 4.3%, so the two spellings differ by a factor of seventeen on that one cell, <i>taŋri</i> costing 1 in 157 against 1 in 2,656 for <i>tanri</i>. The velar form is what these characters wrote; the dental <i>-n</i>- of modern Turkish <i>Tanrı</i> is a later development, and the modern approximation is listed beside every reading in docs/proposed_names_by_price.html rather than here. This row is the file's calibration control. It is the only two-character item whose meaning the Chinese sources state outright, 匈奴謂天為撐犁, “the Xiongnu call heaven <i>chengli</i>”, so its correct reading is not in doubt and its price says what a correct reading costs on this machinery: about 1 in 152, or 8.1% per character. Whole-name prices elsewhere in this file should be read against that figure and not against the single-cell rates quoted in §8.1 of the paper. REASON Taŋri, “heaven, sky, god”. This is the standard identification, which §6.3 of the paper reaches independently by lexicon search: Kāšgārī returns <i>tengri</i> 4th and the <i>Codex Cumanicus</i> 2nd, glossed “sky, heaven” and <i>deus</i>, a thousand years later and with no knowledge of the Chinese record. What this file adds is not the identification and not the vowel, both of which are established, but a measured statement of which reconstructed variant these two characters write. An alternative reading offered in earlier drafts is withdrawn here, and the reason is worth stating because the transcription did not decide it. That reading was <i>tangli</i>, from <i>tang</i> “dawn, light”, and it prices at 1 in 139 against 1 in 152 for <i>tangri</i>. The two differ in one cell, the liquid, where a source <i>l</i> takes a Chinese <i>l</i>- at 97 of 100 and a source <i>r</i> at 185 of 208. That difference is nine percent and it decides nothing. What decides it is the gloss: the Chinese sources say the word means heaven, and <i>tang</i> does not. Where the record states the meaning, the meaning outranks the sound match, which is the same standard §9.1 of the paper applies at 徑路. What is not settled. That <i>tengri</i> is Turkic to begin with, which §6.3 flags: Mongolic has <i>tenger</i>, and Georg derives the Turko-Mongolic word from Proto-Yeniseian *<i>təŋgVr</i>- “high”. Georg 2001 is not among the works WE have been able to obtain, so the vowel claim above has not been checked against it.				
孤塗 gutu	kua da	kua da	Kuti	the blessing of heaven, with the third-person possessive

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 孤 <i>kua</i> → <i>ku</i>: the default velar onset, direct, with the coda open. 塗 <i>da</i> → <i>tɪ</i>: a dental onset carrying the <i>-t</i> of <i>kut</i>, with the vowel free. Two characters, two syllables, nothing inserted and nothing discarded. The frame is kɭɣlh – tɫd and both codas are unwritten, so a source <i>-r</i>, <i>-l</i>, <i>-s</i> or <i>-z</i> could hide in either. 66 attested headwords fit it. REASON <i>kuti</i>, “the blessing of heaven”, with <i>-i</i> as the third-person possessive. <i>kut</i> is the divine favour that legitimises a ruler: Clauson records it in the hendiadys <i>kut kɪv</i> “divine favour”, and it is the word the <i>Kutadgu Bilig</i> is named from. The possessive is not padding but what Turkic requires of the second noun in a genitive-possessive phrase, <i>tengri kuti</i>, “the <i>kut</i> of heaven”, which is the shape of the full title 撐犁孤塗單于. The phrase is attested, not assembled. Clauson’s entry for <i>kut</i> cites the <i>Irk Bitig</i> for <i>tengri kutinta</i> “by the favour of heaven”, which is <i>tengri kuti</i> with a locative on it, and the Orkhon inscriptions for the dative <i>kutiŋa</i>, in the passage where the ruler’s mother is said to have enjoyed the favour of heaven. So both the possessive form and the two-word phrase are on record in Old Turkic, and the Chinese title maps onto them element for element. That is now checkable in the primary edition rather than only in Clauson’s report of it: Tekin’s glossary to the <i>Irk Bitig</i> carries <i>kut</i> as a headword, glossed “divine favor, blessing, good fortune”, and its citations include <i>kutinta</i> at omens 15 and 56, the first of them in the phrase <i>tengri kutinta</i>. The manuscript is ninth-century. In Old Turkic that suffix is <i>-i</i> / <i>-i</i>, so <i>kuti</i> rather than <i>kutu</i>; the <i>-u</i> of modern Turkish <i>kutu</i> comes from labial harmony, a later development. The two-character spelling is itself evidence for the second syllable. Had the word been a bare <i>kut</i>, a single character was available that writes it exactly, 骨 <i>kuət</i>, with the <i>-t</i> on the page, and the record uses that character in 骨都侯. A scribe who had the one-syllable option and wrote two characters was writing two syllables. The vowel of 塗 is <i>a</i> and matches neither <i>ɪ</i> nor <i>u</i> closely, but §7 of the paper puts vowel recovery at 50.0% with the interval straddling one half, so the vowel is the least informative part of the spelling and cannot decide between them. The Chinese gloss is 子 “son”, and this reading does not translate it. That is recorded against the proposal rather than explained away. The word the gloss wants is <i>oġul</i>, and <i>oġul</i> is barred by the writing rather than merely disfavoured: it is vowel-initial while 孤 carries a velar, and in the Sanskrit corpus a source vowel-initial syllable took a zero or glottal onset in 96% of 183 cases and a velar in none of them. Its medial <i>-ġ-</i> has nowhere to sit either, since 塗 is dental. So <i>oġul</i> is not one of the 66 headwords that fit the frame and never was a candidate; the same wall excluded <i>okluk</i> at 胡祿. Of the 66 that do fit, none means “son”: they include <i>kut</i> “heavenly favour, fortune”, <i>kat</i> “layer”, <i>kata</i> “time, occasion”, <i>ketü</i>, <i>Kerti verus</i> and <i>kötü</i>. The Yeniseian rival, as its author now leaves it. Pulleyblank compared the word with Arin <i>bikjäl</i> “son”. In 2000 Vovin held that Arin form to be isolated in the family, the other languages reflecting Proto-Yeniseian <i>*puʔ-b</i> “son” (Ket <i>hiʔp</i>, Yug <i>fiʔp</i>, Kott <i>fup</i>), and found no Altaic etymology either (Vovin 2000, <i>CAJ</i> 44/1, p. 91). He withdrew it three years later: “I would like to take back my suggestion that Arin form is isolated in Yeniseian” (Vovin 2003, Part 2). He then rebuilds the comparison rather than dropping it, reading Pumpokol <i>p-halla</i> “my son” as a cluster with the first-person prefix rather than as an <i>f-</i>, which makes it cognate with the Arin form, and ruling out Pulleyblank’s Ket <i>qal/qaloq</i> instead. The Yeniseian side of this item is therefore stronger than Part 1 alone would suggest, and it is recorded here as he left it. So this is the one item in the record where two independent kinds of evidence point at different words: the writing supports <i>kuti</i> and the gloss asks for a word the writing excludes. The gloss can be accounted for without supposing the informant was confused about Xiongnu. If the title meant “the heaven-favoured chanyu”, the Chinese expression that fits it is 天子, “Son of Heaven”, which is their own imperial title. 天子 has two characters where the Xiongnu title has two words, so the element glosses follow mechanically from that equation: 撐犁 aligns with 天和 孤塗 with 子. On that account the first gloss is a true lexical gloss, since 撐犁 is <i>tengri</i> “heaven”, and the second is a back-formation from the phrase-level equivalence rather than testimony about what 孤塗 means. This is an explanation of the gloss and not a test of it, which is why the reading is offered here rather than argued in the paper.				
屠耆 tuqi	da gɪ [†]	da gɪ	Doġru	straight, upright, honest
PHONETICS 屠 <i>da</i> → <i>do</i>: a dental onset, direct. 耆 <i>gi</i> → <i>ġru</i>: a velar onset, with the <i>-r</i> unwritten, which is forced rather than permitted, since Later Han carries no <i>-r</i> in any coda position. This is one of only three items in the record the Chinese sources gloss, and the gloss is in both of them. 匈奴謂賢曰屠耆, “the Xiongnu call the worthy <i>tuqi</i>”, stands in the <i>Shiji</i> and in the <i>Hanshu</i> alike, and 左屠耆王 is the Wise King of the Left. So this row has what only 撐犁 and 孤塗 otherwise have, a stated meaning to test a reading against. The frame is wide and the gloss is unmet. Seven hundred and forty two-syllable headwords fit these two characters and not one of them means 賢. <i>Bilge</i>, which is the Turkic word for “wise, sage”, prices at 1 in 20,043 and needs 屠 to write a <i>b-</i>, so the word the gloss actually wants is out of reach of the spelling. That is the same shape as 胡祿, where the gloss names a target, a Turkic word exists for it, and the transcription rules the word out. The Shiji carries a textual variant on the first character. Xu Guang notes 屠一作諸, that 屠 is written 諸 in one version, which would give a palatal onset in place of a dental. That branch was searched too and returns no word in the target sense either. The reading proposed here costs 1 in 183,594, 1,208 times the control at 撐犁, and it means “straight” rather than “worthy”. REASON This is the author’s proposal, and the comparison is ours. We have not found this reading proposed in print. The search that would have found it covered every book held for this project, including Golden’s <i>Introduction to the History of the Turkic Peoples</i> in a machine-readable form made for the purpose, and returned nothing. It is recorded here as our own comparison and not as a reading taken from the literature. The Codex Cumanicus has <i>Togru</i> glossed <i>rectus</i>. The Chinese sources gloss 屠耆 as 賢 “worthy, wise”, and it appears in 左右屠耆王, the Wise Kings of the Left and Right. The senses meet more closely than “straight” suggests. Clauson derives <i>toġuru</i>: / <i>toġru</i>: as a gerund of <i>toġur-</i>, physically “straight” and then metaphorically “straight, honest, upright, true”, and that moral sense is old rather than modern: a fourteenth-century glossary renders it with Arabic <i>al-tiqa</i> “trustworthy, honest”, and <i>toġru ayt-</i> with <i>šadaqa</i> “to tell the truth”. The word survives across the whole Turkic family and as modern Turkish <i>doġru</i>, “straight, correct”. Toġrıl and Toġrul are not offered in support. The two names resemble <i>toġru</i> in sound, and they are well attested: Kâşgarî records that men are called <i>Toġrıl</i>, an <i>Alp Toġrul Tégin</i> appears in a Uyghur list of proper names, and the first Seljuk sultan bore the name. But <i>Toġrıl</i> is a bird of prey, and Clauson derives it from <i>toġra-</i> rather than from <i>toġur-</i>, noting that it never survived as a common noun. What makes the two look like one word is a separate homophone, the passive <i>toġrıl-</i> “to be upright”, which is from <i>toġur-</i> and survives as modern Turkish <i>doġrul-</i>. So the names are recorded here to head that resemblance off, not to rest anything on it: the case for <i>Doġru</i> stands on the Chinese gloss 賢 and on the attested moral sense of <i>toġru</i>. The paper does not argue this reading; §9.1 there tests only proposals taken from the literature. What is not settled, and on this row it is a good deal. The price, which is the second most expensive in the file; the gloss, which <i>toġru</i> meets only by the moral extension of “straight” to “upright” and not directly; and the frame, which admits 740 words and so cannot prefer this one. It is kept as an author’s proposal under the standard §S1 sets out, not as a supported reading.				
谷蠡 luli / guli	kok [†] le ^{a†}	kok le	Körklü	glorious, imposing

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 谷 <i>kok</i> → <i>körk</i>: the default velar onset, with the <i>-k</i> coda written and the <i>-r</i>- unwritten before it. Later Han has no <i>-r</i> coda, so a source <i>-rk</i> cannot be written at all without spending a further character on an <i>l</i>- syllable; leaving it unwritten is the only option two characters allow, and it is the majority practice: step 31 counts 63 verified pairs whose source form has a liquid before a consonant, and the liquid is unwritten in 54 of them, 86% (達磨 for <i>dharmā</i>, 薩達摩 for <i>saddharmā</i>, 播鉢吒 for <i>parpaṭa</i>). In 6 of the 63, 10%, the scribe spent a further character carrying an <i>l</i>- onset instead, so the step this reading asks for has a measured minority against it rather than none. 蠡 <i>le^B</i> → <i>lū</i>: the <i>l</i>- onset direct, with an open coda. Two characters, two syllables, one unwritten consonant. Both are polyphonic characters: the table gives 谷 as <i>kok</i>, <i>jok</i> and <i>lok</i>, and 蠡 as <i>le^B</i> and <i>loi^B</i>; this reading takes the first of each. REASON <i>Körklü</i>, “glorious, imposing”, taking the word at the broader end of a range that runs from beauty to splendour. The word is a headword in both lexicons used here, and its sense broadens across the centuries, which the reading has to reckon with. Kāšgarī’s own Arabic is at the narrow end: he glosses <i>körk</i> as <i>ḥusn</i>, <i>jamāl</i>, beauty, and <i>körklüg</i> as <i>jamīl</i>, beautiful, with <i>ḥasan</i> elsewhere. Earlier drafts of this entry gave the gloss as <i>iyi</i>, <i>güzel ve gösterişli</i>, “good, beautiful and imposing”, which is the Türk Dil Kurumu index’s Turkish rendering rather than Kāšgarī’s Arabic, and the “imposing” in it is not his. Clauson’s attestations sit at the same narrow end: <i>körklüg menizliğ</i> “good-looking” in Uyghur Buddhist texts, <i>kuba körklügüğ kör</i> “see the fair-haired beauty” in the <i>Kutadgu Bilig</i>, and a Kipchak <i>inen körklü</i> “extremely beautiful”. The broader sense appears later, and it is well attested. The <i>Codex Cumanicus</i> glosses <i>körk</i> as <i>decus</i>, glory or dignity rather than prettiness. The <i>Book of Dede Korkut</i>, Oghuz and preserved in fifteenth and sixteenth century manuscripts, carries <i>görklü</i> as a fixed devotional epithet: <i>adı görklü Muhammed</i>, “Muhammad whose name is glorious”, recurs as a formula before battle, and a formulaic catalogue runs <i>ayna günü okunanda hutbe görklü, kulak urup dinleyende ümmet görklü</i>, “glorious is the sermon read on Friday, glorious the congregation that hears it”. In the Deli Dumrul story God is addressed as <i>görklü Tanrı</i>. None of these referents is merely pretty, and the highest of them is God. Modern Turkish continues the same end of the range with <i>görkem</i> “splendour, magnificence” and <i>görkemli</i> “imposing”, both from this stem. Two things keep that from settling the row. The broad sense is Oghuz and late, four centuries after Kāšgarī and far later still than the record this supplement reads, and in <i>Dede Korkut</i> it attaches almost entirely to Islamic religious referents, which is a devotional register rather than a general one. The narrow sense is meanwhile still the current one elsewhere in the same century: the fourteenth century Kipchak vocabularies Clauson cites give <i>körk</i> as <i>al-ḥusn wa-l-jamāl</i> (Houtsma 19, 12) and <i>al-ḥusn</i> (Ibn Muḥannā 80), beauty and nothing further. In the oldest layer <i>körk</i> renders Sanskrit <i>rūpa</i>, “form” (Suv. 164, 20), and the related <i>körtle</i> is a component of women’s names in the Uyghur documents. WE therefore record the broad sense as attested and the narrow one as the older, and price this reading on its form rather than on its gloss. Clauson records the form across the whole span, with Kipchak and Coman <i>körklü</i> / <i>körkli</i> and Osmanlı <i>görklü</i>. The Kipchak form matters for the spelling: <i>körklü</i> carries no final <i>-g</i>, so the open coda of 蠡 is simply correct and nothing further needs licensing. The word is used of exalted figures and not only of pretty objects: an Old Turkic Buddhist text has <i>tengridem kız azu tengri oğlanı teg körkle oğlan</i>, “a child as beautiful as a divine girl or a son of God” (TT V 12, 127, via Clauson). Two caveats belong with that citation: Clauson calls the text a fairly late one, and he queries whether the <i>-le</i> form <i>körkle</i> / <i>körtle</i> is native rather than borrowed. The form this reading uses, <i>körklüg</i>, is not in doubt, being a straightforward derivation in <i>-lüg</i> from the attested <i>körk</i>. An authority assigns 谷 a different reading in this very title. Pulleyblank’s <i>Lexicon</i>, at K1202a, glosses 谷 read <i>lu</i> as “luli king, Xiongnu title”, cross-referring <i>gu</i> and <i>yu</i> as the other readings. That is the <i>lok</i> branch, on which no Turkic word can fill the first syllable at all, so if Pulleyblank is right this reading is not available and 谷蠡 is not a Turkic word. The proposal is recorded with that against it, unresolved. The other readings of 谷 do not compete. On <i>lok</i>, which the office’s conventional name, the Luli Kings, would suggest, the first syllable begins with a Chinese <i>l</i>-, and a Chinese <i>l</i>- writes a source <i>l</i>- or <i>r</i>-. Turkic admits neither initially: Clauson calls initial <i>r</i>-, like initial <i>l</i>-, a sound entirely foreign to the language, and loanwords that established themselves took a prosthetic vowel. That branch yields no candidates at all, so if 谷 is <i>lok</i> here, 谷蠡 is not a Turkic word. On <i>jok</i>, which is Schuessler’s palatal glide, the frame is y...k plus <i>l/r</i> and admits <i>yüklüg</i> “laden” and <i>yokla</i>- “to rise, ascend”, neither of which suits an office. What cannot be tested here. The Chinese sources gloss 撐犁 and 孤塗 and say nothing about 谷蠡, so this row carries no semantic constraint at all, only the rank of the office: the 谷蠡王 of the Left and Right stood among the four highest, paired with the Wise Kings. The choice among the candidates therefore rests on the epithet argument that §S1.3 names as this supplement’s main weakness, and the reading is offered as permitted rather than confirmed.				
當戶 danghu	taŋ [†] ga ^B	tang ga	Tamga	the ruler’s seal, and the officer who keeps it
PHONETICS 當 <i>taŋ</i> → <i>tam</i>:- a source <i>t</i> written with a Chinese <i>t</i>- is 55 of 198, 27.8%, and a source <i>a</i> written with an open vowel is 1127 of 1634, 69.0%. The coda is the weak link and is stated rather than buried: the word carries an <i>-m</i> and the character carries the velar <i>-ŋ</i>, a change of place which the corpus permits in 17 of 58 cases, 29.3%, but does not favour. 戶 <i>ga^B</i> → <i>-ga</i>: a source <i>g</i> written with a Chinese <i>g</i>- is 64 of 110, 58.2%, with the vowel open again. The pair is the corpus device for a medial nasal-plus-velar: given a written <i>-ŋ</i> coda followed by a velar onset, the source had a nasal followed by a velar stop in 26 of 29 cases, and the three exceptions are all one word. The whole reading costs 2.26%, about 1 in 44. That is more likely than the 1 in 152 of the control at 撐犁, and the figure is given here so that a reader can weigh it rather than take the reading on trust. What these characters cannot write. A word with a single velar nasal and no velar stop, which is why <i>toŋa</i> stood here in earlier drafts and has been moved to 車牙若鞮, where two characters write it directly. The <i>Codex Cumanicus</i> forms that appear to fit this frame, <i>tengis</i>, <i>tongus</i>, <i>tangis</i>, are medieval Latin spellings of a single velar nasal, as the Diwān’s own <i>teñiz</i> and <i>toñuz</i> show for the same words, and they are excluded for the same reason. REASON <i>Tamga</i>, the seal of the ruler, and by extension the officer who keeps it. 當戶 is an administrative office in the Xiongnu hierarchy rather than a ruler’s name, so the register wanted is court vocabulary, and <i>tamġa</i> is exactly that. The sense is attested, though not under the headword. Kāšgarī’s own entry for <i>tamga</i> gives an unrelated sense, a branch of water or an anchorage, but the derivatives in the same dictionary carry the seal throughout: <i>tamġalamak</i>, “to strike the khan’s seal” (III 353), and <i>tamġalıg</i> and <i>tamġalıq</i>, “objects bearing the khan’s seal” (I 527). Clauson adds an Uyghur civil document, <i>beg tamġası elginde</i>, “a beg’s seal is in your hand” (TT I 129). So the administrative sense is on record in two independent places, and the later Turkic and Mongol chanceries make <i>tamga</i> the ordinary word for the seal, with <i>tamġacı</i> its keeper. What is not settled. The price, 1 in 44, which is worse than the threshold the paper uses elsewhere, and the reader is told so above. The nasal, which is a change of place rather than a direct correspondence. That the seal sense reaches the headword only through its derivatives, so the chain has one link more than at a row where the dictionary glosses the word itself. And whether the Chinese office 當戶 had anything to do with seals, which the histories do not say and this reading assumes.				
閼氏 yanzhi / ezhi	ʔat dze ^{B†}	’at ce	Elti	lady; a woman related by marriage

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 關 <i>ʔat</i> → <i>el</i>⁠: a vowel-initial character whose written -t carries the source liquid. Later Han had no -r or -l in any coda, so a source syllable ending in one was written with a -t or left open: of 37 such syllables in the Sanskrit corpus, 21 were left open and 14 took a -t, 38%. 氏 <i>dže</i>^B → <i>ti</i>⁠: a dental onset with the vowel free. The first character is not polyphonic: Schuessler's table gives 關 the single value <i>ʔat</i>, and Baxter and Sagart give Middle Chinese <i>ʔat</i> from Old Chinese <i>*qʔat</i>. The -n that the parallel spelling 焉支 carries belongs to 焉, a different character, which Baxter and Sagart give two readings, both in -n. The second character is another matter: the table gives 氏 as <i>dže</i>^B, <i>gie</i>^B, <i>tše</i> and <i>kie</i>, and Baxter and Sagart reconstruct <i>*k.deʔ</i> with a velar preinitial, which is where the velar readings come from. This item is character-anchored in the emission row; the second character is less constrained than that alone suggests. REASON <i>Elti</i>, “lady”. Modern Turkish <i>elti</i> is what each of two brothers' wives calls the other, and in dialect what two wives of one man call each other. It is not only modern: Clauson records <i>élti</i>: glossed “lady”, standing opposite <i>al-jāriya</i> “slave girl”, in a fourteenth-century Arabic-Turkish glossary, which is the sense a consort's title would want. Nişanyan takes Middle Turkic <i>élti</i> “married woman, as a respectful term” and derives it, with a stated doubt, from Old Turkic <i>élt</i>- “to carry, to lead”. If that holds, the word has the shape of <i>kelin</i> “bride”, which Clauson derives from <i>kel</i>- as “one who comes in (to the family)”: a deverbal noun naming a woman by how she entered it. 關氏 is the title of the chanyu's principal consort. Why this replaces the earlier proposal in this supplement. The earlier reading was <i>ece</i> “elder sister”, later <i>eci</i> “senior woman, lady”, both of which are in the Kâşğarî index (I, 86 and I, 87) and the second of which is the better semantic fit. Both discard the written -t of 關, and the corpus makes that expensive: where the source has a single affricate or sibilant between two vowels, the preceding character was open in 134 of 143 junctions, 93.7%, and carried a -t in 2, 1.4%. That is a worse cell than the 1 of 37 the paper uses in §8.1 against a rival reading elsewhere, and the two are comparable because both are single cells, so consistency required either conceding it or finding a reading that uses the -t. <i>elti</i> uses it, at 38%. The earlier proposal is recorded here rather than removed, since the reasoning that led away from it is the useful part. What is not settled. <i>elti</i> is not in the Türk Dil Kurumu index nor in the Codex Cumanicus, and the fourteenth-century attestation is in the edition Clauson marks “(only)”. Pulleyblank read the title as Turkic and Mongolian <i>qatun</i>; Vovin rejects that and proposes instead a Yeniseian <i>*alte</i> or <i>*elte</i> “wife”, from Proto-Yeniseian <i>*ʔal(V)it</i>- (Kott <i>alit</i>, Assan <i>alit</i>, Arin <i>biqam-alte</i>), noting that it “could be transcribed in Chinese only as <i>*ʔat-tej</i> or <i>*ʔen-tej</i>”. He is right about the Chinese and the same spelling writes <i>elti</i> by the same steps, so the transcription does not choose between the two languages. §9.1 of the paper sets that out as a measured case of the negative result of §6.1.				
骨都侯 guduhou	kuət ta go	kuit ta go	Kutak	the blessed one
PHONETICS 骨 <i>kuət</i> → <i>kut</i>: default velar onset with the -t coda written, the tightest single syllable in this supplement. 都 <i>ta</i> then begins with the same dental, and Table 8 of the paper measures what that means: at a junction where a written coda is followed by the same-place onset, the source word carries that consonant once rather than twice in 121 of 140 cases (86%). So 骨都 transcribes <i>kuta</i>, not <i>kutta</i>. 侯 <i>go</i> → <i>-k</i> closes it, a Later Han voiced g- standing for a source k more often than for a source g. REASON <i>kut</i> is “kut, uğur, devlet, baht, talih, saadet” in Kâşğarî (heavenly favour, fortune, sovereign grace) and Clauson records the verb <i>kutad</i>- “to be blessed”, the stem behind the <i>Kutadgu Bilig</i>. <i>Kutak</i> itself is not attested: it would be a deverbal or diminutive formation in -(A)k on that stem, which is a live suffix but not one witnessed on this word. Compare <i>Qutalmış</i>, ancestor of the Seljuks of Rum. Note also that <i>kut</i> is the Turkic candidate for 孤塗, where the Chinese gloss 子 “son” fits the Yeniseian proposal better and not this one.				
且渠 juqu	tsa ^B ɡia	tsa ɡia	Çaka	the striker; from çak- ‘to strike fire, to strike a blow’
PHONETICS 且 <i>tsa</i>^B → <i>ça</i>: ts- for a source affricate is a minority spelling but an attested one, 15% in the Sanskrit corpus and 9% in Ligeti, against a tś- character as the default at 37%. 渠 <i>ɡia</i> → <i>ka</i>: a Later Han voiced g- stands for a foreign k more often than for a foreign g. Both steps are permitted by the measured correspondences. REASON <i>çak</i>- “to strike fire from flint, to sting, to slander”, which Clauson describes as onomatopoeic in character and of violent action, physical or mental. No <i>çaka</i> headword exists in Kâşğarî, the Codex or Clauson, only the verb and its derivatives <i>çakmak</i>, <i>çakır</i>, <i>çakılmak</i>, so <i>Çaka</i> would have to be the -A converb of that stem, and a bare converb serving as an office title is not a demonstrable pattern. The 11th-century emir Çaka Bey of Smyrna (Byzantine Τζαχάκ, <i>fl.</i> 1081–93) shows that <i>çak</i>- could yield a personal name, but that is a personal name set against a title. The Chinese sources do not record what the office of 且渠 did, so no sense can be tested against a gloss.				
徑路 jinglu	keŋ ˢ laˢ	keng la	Kıŋrak	a large cleaver-like knife
PHONETICS 徑 <i>keŋ</i>^ˢ → <i>king</i>: default velar onset with the -ŋ coda written. 路 <i>la</i>^ˢ → <i>ra</i>: a Chinese l- for a source r-, the direction those counts show in 12 of 14 attestations. The Turkic word's final -k is not written, which Table 7 of the paper licenses at 30%. REASON Not our reading. Friedrich Hirth compared 徑路 with Turkic <i>kıŋrak</i> in <i>The Ancient History of China to the End of the Chóu Dynasty</i> (1908), pp. 65–70, calling it the oldest Turkish word on record. Kâşğarî glosses <i>kıŋrak</i> as a knife like a butcher's cleaver, for cutting meat and dough, and the Chinese sources describe 徑路 as a blade used in oath-taking, so gloss and referent agree across unconnected sources. Turkic origin, on Clauson's analysis. Clauson derives <i>kıŋra:k</i> as a deverbal noun from an unattested <i>*kıŋra</i>⁠:, denominal from <i>kıŋır</i>, in the sense of something curved or something that cuts crookedly, so the word is motivated inside Turkic by the standard authority rather than by us. Kâşğarî's index carries the same family (<i>kıŋru</i> “askew”, <i>kaŋrak</i> “palate”, <i>koŋur</i>- “to wrench, bend back”), and the Codex glosses <i>Kingir</i> as <i>curvus</i>. Clauson also records the word alive in Teleut as <i>kıŋırak</i> and Kirghiz <i>kıŋarak</i> / <i>kıŋırak</i>, a rough two-edged knife for cutting felt and scraping hides. Two cautions. Kâşğarî's text at III 382 reads <i>kırpa:k</i>, corrected in the margin to <i>kıŋra:k</i>, so the form rests on an emendation; and of the family it is <i>kıŋır</i> that fits the transcription exactly, while <i>kıŋrak</i>, the member meaning a knife, needs the unwritten final -k allowed at 30%. Iranian origin. The object has also been derived from an Iranian word for a short sword, the <i>akinakes</i> type, historically plausible, since the Xiongnu's western neighbors were Iranian-speaking. But 路 is a liquid-onset character, so the channel requires a liquid in second position, and the akinakes frame (k–n–k) has none. As stated, that account fails on the frame. Two cautions there too: we have not seen the exact Iranian etymon its proponents advance, and the channel shows what a transcription permits, never what it requires. §9.1 of the paper sets both origins out.				
服匿 funi	buk [†] ɳɛk	buk nɛk	Bukaç	a clay jar or cooking pot

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS Where the Chinese gloss comes from, since it is not in the Xiongnu chapters. 服匿 does not occur in <i>Shiji</i> 110, <i>Hanshu</i> 94 or <i>Hou Hanshu</i> 119; every 匿 in those chapters is the ordinary verb “to hide”. The term is in <i>Hanshu</i> 54, the life of Su Wu, where the king gives him 馬畜服匿穹廬, horses, livestock, a <i>funi</i> and a felt tent. Meng Kang, quoted by Yan Shigu, describes the object as resembling a jar, 罍, with a small mouth, a large belly and a flat bottom, used for wine and dairy. What the two characters can write. 服 gives a <i>b</i>- onset, which writes a source <i>b</i>, <i>v</i> or <i>p</i> and nothing else, with a rounded nucleus, which takes a Turkic <i>o</i> or <i>u</i> at 63.9% and a Turkic <i>ö</i> or <i>ü</i> at 96.9%, and a written <i>-k</i>. 匿 <i>ník</i> gives a nasal onset, which writes a source <i>n</i> at 28.2%, a <i>d</i> at 3.8% and a <i>y</i> at 2.8%, with a central nucleus, the least informative class in the table, and a second written <i>-k</i>. A vowel-initial second syllable is not available, at 1 of 140. 服 <i>buk</i> → <i>bu</i>, with its velar <i>-k</i> carrying the <i>k</i>- of the following syllable, which is the compression device at 93 of 1,110, 8.4%. 匿 is then left writing nothing, at 152 of 1,110, 13.7%, and that is the objection to this reading rather than a footnote. The two devices together cost more than all the phoneme rates in the row, and the whole reading comes to 1 in 237, against the 1 in 152 of the control at 撐犁. REASON Bukaç, a clay jar or cooking pot. Kâşğarî glosses it <i>su kabı</i>, <i>topraktan yapılan çömlek</i>, a water vessel and an earthenware pot, and Clauson records it twice from him, <i>al-ṭaūr wa-l-qidr min xazaf</i>, “a jar or cooking pot of clay”, at I 357 and I 411, where it appears in the pair <i>eşiç bukaç</i>. The Chinese description and the Turkic gloss are the same object, arrived at independently. Meng Kang’s small-mouthed, big-bellied earthenware vessel for liquids and Kâşğarî’s earthenware water pot were written six centuries apart with no contact between them, and \$9.1 of the paper treats that kind of agreement as stronger than a sound match. Why the reading given here in earlier drafts is withdrawn. That form is not a headword in Kâşğarî’s index, the <i>Codex Cumanicus</i>, Clauson or the <i>Irk Bitig</i>. The frame search returns fourteen candidates for this row and none of them is a container: <i>bakanak</i>, <i>bakanuk</i>, <i>bakan</i>, <i>bagna</i>, <i>bekni</i>, <i>benek</i>, <i>bogun</i>, <i>büken</i>, <i>bükin</i>, <i>bükün</i>, <i>ben</i>, <i>bün</i> and two from the Codex. The withdrawn form was the Later Han syllables respelled in Turkish letters with the Chinese meaning attached, which is the failure \$6.3 of the paper measures. The best fit on sound alone is a different word and does not mean this. <i>Bük</i>, “corner, nook” and “dense woodland” in Kâşğarî at I 333 and I 245, prices at 1 in 27, the lowest figure on this row, because a Turkic front rounded vowel takes a rounded Chinese character 58 times in 60. It leaves 匿 idle in the same way and it is not a vessel, so the row is not read on it. What is not settled. That one of two characters does no work, which is the same objection recorded at 虛閭權渠. And whether the word is Xiongnu at all: in the same commentary Jin Zhuo reports that people north of Hedong called a small two-<i>dou</i> stone vessel a <i>funi</i>, which would make it a Chinese regional term rather than a borrowing, and nothing here decides between the two.				
甌脫	ʔo t ^h uas	’o tuat	Otag	a tent or shelter, hence a frontier post
outuo PHONETICS 甌 ʔo → o: vowel-initial on a glottal character, which is what a vowel-initial syllable takes 137 times in 140, 97.9%, and the strongest single cell in the table. 脫 → <i>tag</i>: the character has three Later Han values and this reading takes <i>t^huas</i>, the one without a stop coda, so the <i>-g</i> of the Turkic word goes unwritten, which a final velar does in 9 of 23, 39.1%. Two characters, two syllables, nothing inserted. The whole reading costs 1 in 71, well inside the 1 in 152 of the control at 撐犁. The reading given here in earlier drafts inserted a medial -y- and cannot be priced at all, because it asks three syllables of two characters and neither carries a coda that would license the compression. What the Chinese sources say the thing is. They say it three times, and all three commentators define it by watching. The <i>Shiji</i> has 各居其邊為甌脫, each dwelt at its own border as <i>outuo</i>, and Wei Zhao glosses it 界上屯守處, a place of garrison and watch on the boundary. In the <i>Hanshu</i> Fu Qian has 甌脫作土室以伺也, that an <i>outuo</i> is an earth chamber made in order to watch from, and Yan Shigu 境上候望之處若今之伏宿舍也, a place of watching on the frontier like the hidden billets of his own day; he also gives the readings, 甌音一侯反 and 脫音土活反. The word is not only a structure: the <i>Hanshu</i> reports a 甌脫王, an <i>outuo</i> king taken alive, and 發人民屯甌脫, people sent to garrison the <i>outuo</i>, and 甌脫外棄地, the abandoned land beyond it. So it is a post, an office and a frontier zone at once. REASON Otag, a tent or shelter. Kâşğarî has <i>otag</i> at III 208, and Clauson follows the word across the family: Xwarazmian fourteenth century <i>otaġ</i> “tent”, Chagatay <i>otaġ</i> / <i>otak</i> “dwelling house”, and the same word shortened in Ottoman to <i>oda</i>, which is the ordinary modern Turkish word for a room. A frontier post called the shelter is what Fu Qian’s earth chamber and Yan Shigu’s hidden billet describe. A cheaper word exists and is not taken. <i>Udug</i>, which Kâşğarî glosses <i>uyanık</i>, awake or alert, at I 63 in the phrase <i>udug köñüllüg er</i>, a man of alert mind, prices at 1 in 26, nearly three times better. It matches what the commentators say the place is <i>for</i> rather than what it is, and it is an adjective: naming a post “awake” needs a nominalising step the word does not carry, and the same dictionary’s <i>udugluk</i> means inattention, so the family is not settled on the sense either. WE record it as the cheaper reading and take the better-formed one, and a reader who prefers <i>udug</i> is not contradicted by anything measured here. What is not settled. Neither word means “a place of watching”, which is what the gloss actually says; <i>otag</i> supplies the structure and <i>udug</i> the function, and the record wants both.				
胡祿	go [†] lok	go lok	Külüg	famous, renowned
hulu PHONETICS 胡 <i>go</i> → <i>kü</i>: the devoicing is normal, a Later Han voiced g- standing for a source k more often than for a source g; ü cannot be written and surfaces as o. 祿 <i>lok</i> → <i>lüg</i>: the <i>-k</i> coda is written. The consonant frame k–l–k is secure and only the vowels are free. REASON <i>külüg</i> “famous, renowned” is an Old Turkic epithet of praise, recorded by Clauson; note that the form is <i>külüg</i>, not <i>külük</i>, and it is not a headword in either lexicon used here. Two objections stand. First, the Chinese-side record gives the meaning of 胡祿 as “quiver”, not a term of praise, though that gloss is itself marked low-confidence and chronologically shaky. Second, the reading the gloss wants is barred by the transcription: the one Turkic quiver word whose skeleton fits is <i>okluk</i> (Kâşğarî, glossed <i>sadak</i>), and it is vowel-initial, whereas 胡 carries a voiced velar. In the Sanskrit corpus a source vowel-initial syllable was written with a zero or glottal onset in 96% of 183 cases, and with a velar in none. So this is an item where the Chinese gloss names a target, a Turkic word exists for it, and the spelling rules that word out, recorded because a negative of that shape is worth more than a resemblance.				

TABLE S5 The titles and the handful of common nouns the Chinese histories preserve, in the order of Appendix A. The lexical items are the more interesting group because some of them are glossed in the source, which means a proposal about them can in principle be checked.

S1.10 Ethnonyms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
匈奴 Xiongnu	huoŋ na	huong na	Huŋ	the ethnonym, from the first character alone
PHONETICS 匈 <i>huoŋ</i> → <i>hung</i> : direct, with the -ŋ coda written. 奴 <i>na</i> is discarded rather than read, on the assumption that 奴 “slave” is a Chinese pejorative appended to the name rather than part of it. REASON Dropping 奴 is common in the literature and what is proposed is only the vocalisation of the remaining syllable.				
羯 Jie	kiat	kiat	Kır	the open steppe; those who live on it
PHONETICS A single character, <i>kiat</i> . The onset is the default: a Chinese <i>k</i> - writes a source <i>k</i> in 204 of 241 cases, 84.6%, and a source <i>g</i> in 19, so the first consonant is a velar and little else. Schuessler gives the nucleus two classes, open and central. The written -t is where the frame opens out, and an earlier draft of this entry did not allow for it. Later Han has neither a liquid nor a sibilant coda, so a source <i>-l</i> , <i>-r</i> , <i>-s</i> , <i>-ʃ</i> or <i>-z</i> had to be written with a stop or left off the page, and the dental stop is the usual choice. This character's <i>-t</i> therefore writes a Turkic <i>-t</i> or <i>-d</i> at 53.4%, a <i>-p</i> or <i>-b</i> at 60.0%, a liquid at 46.2%, an <i>-m</i> at 30.4%, a sibilant at 20.0%, a <i>-ŋ</i> at 16.7% and a velar at 13.0%. Seventy-six single-syllable headwords in the three lexicons fit the frame, not two. Thirty of them price at 1 in 13 or better: <i>kat</i> “layer” and <i>kar</i> “snow” at 1 in 3, <i>kal</i> “old man” at 1 in 3, <i>kam</i> “shaman” at 1 in 5, <i>kut</i> “fortune” at 1 in 7, <i>kul</i> “slave”, <i>kol</i> “arm”, <i>kaʃ</i> “jade”, <i>kaz</i> “goose” and <i>kas</i> “bark” at 1 in 8, <i>kum</i> “sand” at 1 in 12, <i>kır</i> at 1 in 39. One of them is an ethnonym of the ordinary kind. Kāşgarî glosses <i>kır</i> as <i>kır</i> , <i>basık dağ</i> , <i>açık yer</i> , open country, low hill, open ground, and a people named from the terrain it lives on is as common a formation as a people named from a self-designation. It reaches the frame by the route this row turns on: Later Han has no liquid coda, so a source <i>-r</i> was written with a dental stop 16 of 43 times, 37.2%, or left off. It is offered as permitted and not as preferred, and this row cannot prefer anything. One character with a common onset and a common coda constrains almost nothing, which is the situation §6.3 of the paper measures as chance, and the two readings actually on the table are inside each other's noise: <i>ket</i> costs 1 in 34 and <i>kır</i> 1 in 39. The one thing the spelling does say is about the vowel, and step 53 counts it. Characters whose Schuessler vowel contains <i>ia</i> write a source <i>a</i> in 28 of 28 cases; a source <i>e</i> is 2.1% of the corpus and appears in none of them. That measurement is recorded here and deliberately not used as an argument. The Jie appear in the fourth century CE and the corpus behind the figure is Later Han, which is the period mixing §7 of the paper warns against; and Proto-Yeniseian <i>*keʔt</i> is a reconstruction of a proto-form, so it does not fix the vowel of the fourth-century word and the measurement is not aimed at what Vovin actually claims. REASON The character may not be a transcription at all. Two accounts of the ethnonym are in circulation and they cannot both be right. On one, the name comes from 羯室, a place in Shangdang commandery in what is now Yushe County, Shanxi, where the group settled; on that account 羯 is a Chinese place name and no search over its sounds can mean anything, because there is no foreign word behind it. On the other, Vovin's, it transcribes the group's own name for itself, Yeniseian “person”, Ket <i>keʔt</i> , Pumpokol <i>ket</i> . The <i>Jin Shu</i> makes Shi Le, who founded the Later Zhao in 319, a Jie of Wuxiang in Shangdang and derives the group from a branch tribe of the Xiongnu, which fits either account. Nothing in this file decides between them, and the first one would make this row unreadable in principle rather than in practice. Kır, “the open steppe”, and those who live on it. Kāşgarî glosses <i>kır</i> as <i>kır</i> , <i>basık dağ</i> , <i>açık yer</i> , open country, low hill, open ground, and a people named from its terrain is as ordinary a formation as a people named from a self-designation. It costs 1 in 39. It is proposed under this supplement's standard and not as a result. These entries record what the transcription permits, and this one permits <i>kır</i> ; nothing here shows it is right. Seventy-six lexicon words sit on the frame and a dozen price better than the control at 撐犁, so a row that admits that many candidates cannot support a choice among them, and picking the one that suits the thesis is the selection bias §6.3 measures. WE state that plainly rather than leave a reader to notice it. The reading given here in earlier drafts is withdrawn. It was a two-syllable form, and this character carries one syllable: the alignment fails outright, before any question of sense. That form is also not a headword in the Kāşgarî index, the Codex or the <i>Irk Bitig</i> glossary, and the root it would have been built on means the opposite of what was claimed for it. Kāşgarî's <i>kıy</i> - family is <i>kıymak</i> “sözden dönmek; kıymak, eğrilemesine doğramak”, to break one's word and to cut crosswise, with <i>kıyık</i> “cayma, caymak” and <i>kıyturmak</i> “iğrilemesine kestirmek”. Nothing in it is about water, and the sense offered had no source. The reading in print stands unopposed, and it prices well. On these rates <i>ket</i> costs 1 in 34, better than the 1 in 152 of the control at 撐犁, so the transcription does not argue against it. Pulleyblank and Vovin read the ethnonym as <i>*ket</i> and connect it with Proto-Yeniseian <i>*keʔt</i> “person, human being”, the Yeniseian self-designation, reflected in Ket <i>keʔt</i> , Yug <i>keʔt</i> , Kott <i>hit</i> , Arin <i>kit/qit</i> and Pumpokol <i>kit</i> (Vovin 2000, CAJ 44/1, p. 91). The vowel count is not used against it, and now less than ever. The 28 of 28 measurement in the phonetics field would tell against <i>*ket</i> , and with a Turkic reading standing beside it this author has more interest in its telling against <i>*ket</i> , not less. It is therefore stated there and not deployed here, for the two reasons given: the period is wrong, and a proto-form does not fix the vowel of a fourth-century word. Nothing separates the two readings on the sounds. 1 in 34 against 1 in 39 is a difference of nothing, and the reading in print has what this one does not, a proposer and a family of cognates behind it. A reader who takes <i>*ket</i> and leaves <i>kır</i> is not contradicted by anything measured here.				

TABLE S6 Group names rather than personal names, and subject to a further difficulty: a Chinese exonym need not transcribe anything the group called itself.

S1.11 Characters with more than one Later Han reading

CHARACTER	READING PRINTED	OTHER READINGS IN THE TABLE	ITEMS IN WHICH IT APPEARS
鞬	te	de	且鞮侯 壹衍鞬 握衍胸鞬 若鞮 復株累若鞮 搜諧若鞮 車牙若鞮 烏珠留若鞮 烏累若鞮 呼都而尸道臯若鞮 鞬鞮
犁	lei	li	响犁湖 撐犁孤塗單于 撐犁
且	tsa ^B	ts ^h ia ^B · ts ^h ia · tsia	且鞮侯 且渠
呼	ha	ha ^C	呼韓邪 呼都而尸道臯若鞮
單	dʒan	tan	單于 撐犁孤塗單于
累	lui	lyai ^B · lyai ^C · luai ^B	復株累若鞮 烏累若鞮

CHARACTER	READING PRINTED	OTHER READINGS IN THE TABLE	ITEMS IN WHICH IT APPEARS
衍	jan ^B	jan ^C	壺衍鞮 握衍胸鞮
冒	mək	mou ^C	冒頓
胸	huo	huo ^C	胸犁湖
復	buk	bu ^C	復株累若鞮
搜	so	ʃu	搜諧若鞮
支	tʃe	kie	鄴支
曼	man ^C	man · muan ^C	頭曼
服	buk	bu ^B	服匿
氏	dʒe ^B	gie ^B · tʃe · kie	閼氏
當	taŋ	taŋ ^C	當戶
稽	k ^h ei ^B	kei	稽粥
耆	gɿ	tʃi ^B · ki ^B	屠耆
而	ńə	ńɿ	呼都而尸道舉若鞮
胡	gɔ	ga	胡祿
臣	dʒin	gin	軍臣
蠡	le ^B	loi ^B	谷蠡
谷	kok	jok · lok	谷蠡
車	tʃ ^h a	kia	車牙若鞮
連	lian ^B	lian	虛連題
道	dou ^B	dou ^C	呼都而尸道舉若鞮
邪	ja	zia · zia	呼韓邪
頓	tuən ^C	tuət	冒頓
題	de	de ^C	虛連題

TABLE S7 Every character in the record for which Schuessler's table gives more than one Later Han reading, with the reading printed in Appendix A of the paper and the alternatives not used. Generated from [data/derived/polyphony.csv](#), which the numbered script step 24 writes, so this table cannot drift from the repository. The daggers in the tables above mark these same polyphonic characters. Where an alternative would change a proposed reading, the entry for that item says so.

S1.12 The two reconstructions of each character compared

CHARACTER	LATER HAN (SCHUESSLER)	OLD CHINESE (BAXTER & SAGART)	ONSET CLASS	CODA CLASS
粥	tʃuk	t-quk	palatal → velar	—
軍	kun	[k]ʷər	—	dental nasal → liquid
臣	dʒin	[g]i[ŋ]	palatal → velar	dental nasal → velar nasal
稚	ɿ	lɾəj-s	dental → liquid	—
維	wi	g ^w ij	glide → velar	—
詹	tʃam	[t]am	palatal → dental	—
衍	jan	N-q(r)an?	palatal → velar	—
權	gyan	[g]ʷrar	—	dental nasal → liquid

CHARACTER	LATER HAN (SCHUESSLER)	OLD CHINESE (BAXTER & SAGART)	ONSET CLASS	CODA CLASS
韓	gan	[g]ˈar	—	dental nasal → liquid
邪	ja	sə.ɣA	palatal → velar	—
支	tʃe	ke	palatal → velar	—
若	ńak	nak	palatal → dental	—
車	tʃʰa	[t.qʰ](r)A	palatal → velar	—
而	ńə	nə	palatal → dental	—
尸	ʃi	ʃ[ə]j	palatal → liquid	—
道	dou	ʎˈuʔ-s	dental → liquid	—
單	dʒan	dar	palatal → dental	dental nasal → liquid
于	wa	ɣʷ(r)a	glide → velar	—
塗	da	ʎˈa	dental → liquid	—
闕	ʔat	qˈat	glottal → velar	—
氏	dʒe	k.deʔ	palatal → dental	—
路	ʎa	Cə.rˈak-s	—	open → velar stop

TABLE S8 The 22 characters of the record, of the 60 present in both reconstructions, where the two give a different class of segment in at least one position; the other 38 agree. This is the measurement §11 of the paper reports, at 37%. It is not a correction to Appendix A: Baxter and Sagart reconstruct a stage several centuries before the transcriptions and Schuessler one several centuries after, so the two bracket the moment of writing rather than dating it, and this table is the width of that bracket. Their *-s and *-ʔ are treated as having become tones by the Han and are not counted as codas, and a trailing *-j or *-w is read as the offglide of a diphthong rather than as a consonant. Generated from `data/derived/period_depth.csv`, which the numbered script step 43 writes.

S1.13 Every reading priced, most expensive first

FORM, AS WRITTEN IN THE HISTORIES	READING PROPOSED	PRICED AS	CHARACTERS PRICED	PRICE, 1 IN	PER CHARACTER	SENSE
屠耆	Doğru	—	2	188,583	0.2%	straight, upright, honest
且鞮侯	Çatıgka	—	3	224,966	1.6%	one ready for the clash
若鞮	İnakt	Nakt	2	2,217	2.1%	the faithful ones
羯	Kır	—	1	42	2.4%	the open steppe; those who live on it
闕氏	Elti	—	2	1,339	2.7%	lady; a woman related by marriage
冒頓	Böğü Tuğ	Bö Tuğ	2	1,153	2.9%	sage standard
單于	Çarvuş	—	2	845	3.4%	army commander; the earlier form of çavuş
且渠	Çaka	—	2	563	4.2%	the striker; from çak- 'to strike fire, to strike a blow'
撐犁孤塗單于	Tanrı Kutı Çarvuş	Tanrı Kutı Çarvuş	6	21,377,362	6.0%	Heaven-Fortune Chanyu, the full ceremonial title
搜諧若鞮	Sarıg İnakt	Sarg	2	275	6.0%	the yellow, the fair, of the İnakt
服匿	Bukaç	—	2	238	6.5%	a clay jar or cooking pot
孤塗	Kutı	—	2	161	7.9%	the blessing of heaven, with the third-person possessive

FORM, AS WRITTEN IN THE HISTORIES	READING PROPOSED	PRICED AS	CHARACTERS PRICED	PRICE, 1 IN	PER CHARACTER	SENSE
撐犁	Tanrı	Taŋrı	2	157	8.0%	heaven, sky, god
攣鞬	-rındaş	—	2	144	8.3%	the last two syllables of Karındaş, the first lost
响犁湖	Kalık	—	3	1,696	8.4%	sky, heaven
烏珠留若鞬	Çal İnakt	Çal	3	1,139	9.6%	the grey or dappled one, of the İnakt
甌脫	Otag	—	2	73	11.7%	a tent or shelter, hence a frontier post
骨都侯	Kutak	—	3	567	12.1%	the blessed one
呼都而尸道皋若鞬	Kartay Bedük İnakt	Kartay Bedük	6	243,285	12.7%	the great elder, with the title
壺衍鞬	Kayantı	—	3	345	14.3%	avalanche
當戶	Tamga	—	2	47	14.6%	the ruler's seal, and the officer who keeps it
胡祿	Külüg	—	2	42	15.5%	famous, renowned
握衍胸鞬	Okyan Kutı	—	4	1,252	16.8%	the fortune of the arrow flank
虛閭權渠	Kalkan	—	4	1,164	17.1%	shield
狐鹿姑	Kurluga	—	3	128	19.8%	the binder, the encloser
復株累若鞬	Büktel İnakt	Büktel	3	114	20.6%	Büktel, of the İnakt
匈奴	Hung	Huŋ	2	23	20.8%	the ethnonym, from the first character alone
車牙若鞬	Tunga İnakt	Tuŋa	2	22	21.2%	the tiger, or the hero, of the İnakt
烏維	Abı	—	2	17	24.2%	one who hunts or stalks
伊稚斜	Idiya	—	3	66	24.7%	master
烏累若鞬	Uluğ İnakt	Uluğ	2	14	27.1%	the Great, of the İnakt
虛連題	Karındaş	—	3	25	34.3%	kinsmen, those of one womb
徑路	Kıŋrak	Kıŋrak	2	8	34.6%	a large cleaver-like knife
谷蠡	Körklü	—	2	8	35.0%	glorious, imposing
詹師廬	Çamşılı	—	3	17	38.8%	dazzling, radiant
呼韓邪	Kaganya	—	3	16	39.8%	kagan of the flank
郅支	Çitçi	—	2	5	43.4%	border-maker, one who carves the boundary
稽粥	Kiçük	—	2	4	47.5%	the younger
頭曼	Duman	—	2	4	50.7%	dust cloud
軍臣	Köngen	—	2	4	52.2%	the one who is always upright

TABLE S9 All 40 rows priced by the one engine. **Priced as** names the form the figure was taken on and is filled in only where that is not the reading exactly as printed. The six names carrying 若鞬 are priced without it, since §9 of the paper argues that element once and it is priced once, in its own row. 撐犁 and 撐犁孤塗單于 are priced on the phonetic form: 撐 carries a velar nasal, which it writes at 72.7% against 4.3% for a dental *n*, so the two spellings differ by a factor of seventeen on that one cell. 若鞬, 攣鞬 若鞬 and 冒頓 are priced on what their characters write, the reading containing something they do not. **The raw price is not comparable across rows**, because it

falls with every extra character and so sorts by length; the per-character figure is the geometric mean, and the sort is on it. The control at 撐犁 stands at 8.0%, and 27 rows reach or beat it while 13 fall short. Generated from `data/derived/author_proposals.csv` by the same `src/price_engine.py` the entries above use.

S1.14 How to disagree

The repository includes `src/lookup_lexicons.py`, which searches Kāşgarî's *Dîwān*, the *Codex Cumanicus* and any EPUB lexicon placed beside them, folding the spelling conventions that separate Turcological from Turkish orthography (q ≡ k, ñ ≡ ng, ş ≡ s, ı ≡ i). Type any proposed form into it and see what the medieval sources actually have. That is how the *Codex* attestation for *inak* was found, and how the -tI count that ruled out the competing suffix was made.

For the reader who disagrees with our proposed names, we suggest disagreeing not with "this is unlikely", since most of them are, but with "this word is not attested before date X", or "this suffix does not attach to that stem", or "the Chinese character cannot carry that coda". Those are the objections that would be useful for the author.

S1.15 Status

Thirty-five of the forty readings in Tables S2 to S6 carry the same label in the machine-readable file: *author's proposal*; the paper measures the steps it takes for the name but does not present it as a finding. That label is the honest summary of them, and it is not weakened by the fact that some are more attractive than others. Five rows carry a different status, and §S1.1 names them: 若鞮 as *Īnakt*, which the paper argues and which meets its standard; 撐犁 as *Taŋrı*, the standard identification and not ours, which serves as this file's calibration control because the Chinese sources state its meaning outright; 徑路 as *Kıŋrak* and 屠耆 as *Doğru*, two comparisons taken from the literature rather than made here, which the paper then tests; and one compound that is read through its constituent rows.

Twelve rows changed status while this supplement was being written, and the changes are recorded here rather than quietly absorbed. 孤塗 and 谷蠡 had no proposal at all in earlier drafts and now carry one. 冒頓 was argued in the paper in an earlier draft and is now an author's proposal like the others: §9.2 of the paper prices the three values its second character can take and does not choose among them, and the proposal made here is set out in S1.4 with its objections. 屠耆 was carried as a comparison taken from the literature and tested in the paper, and is now an author's proposal like the others: the reading could not be traced to any published source, and the paper no longer argues it. 閼氏, 單于 and 軍臣 carry different readings from the ones earlier drafts gave them, in each case because a measurement made since told against the earlier reading; the entries state what the measurement was. At 軍臣 it was that a source voiceless affricate written with a voiced *dž* - is 0 of 53, which retired *Künçin* and sent the reading to the second of the character's two tabulated values. At 狐婁姑 the reading itself is unchanged, but two objections the entry raised against it are withdrawn: leaving a liquid unwritten before an open character is the majority spelling and not an impossibility, and a written coda reaching forward to the next onset is the commonest use of such a coda. Both had been judged by ear before they were counted. At 羯 the proposal was withdrawn outright: the form was two syllables on a one-syllable character, it is not a headword anywhere, and the root it needed means to cut and to break one's word, so the sense offered had no source. A different reading, *Kır*, was proposed for that row later and is set out there; it is offered as permitted rather than as preferred, since the frame admits seventy-six lexicon words and the reading in print sits inside the same noise. §S1 records why a measurement that would have told against the reading in print is reported there but not used. At 呼都而尸道皋若鞮 the proposal was withdrawn too, and for measurements rather than for want of a source: a source vowel-initial syllable takes a Chinese glottal onset 137 of 140 and a velar 0 of 140, and both elements of the reading sat on velar characters; no character in the name carries a liquid, where a source l- takes a Chinese l- 262 of 288; and the reading left four of the eight characters carrying nothing, where no pair in the 1,110-pair Sanskrit corpus uses more than one character beyond its source syllables. *Uluğ* keeps its place at 烏累若鞮, which has a glottal character for the vowel-initial syllable and a liquid character for the -l-. That is the point of keeping the file: the same word was proposed in two rows, and the transcription, not the sense, said which one could hold it. That row is no longer empty. A reading in which every one of its six characters carries sound, *Kartay Bedük*, was found afterwards and is set out there with its price and with the chronological objection against it; the withdrawal is what made room for it. At 車牙若鞮 it was that a source vowel-initial syllable written with a Chinese ɳ - is 0 of 140, which retired *Çöge-Ok*; the reading that replaces it, *Tuŋa*, was carried at 當戶 in earlier drafts and is moved here because those characters write a nasal followed by a velar stop and this word has a single nasal. 當戶 carries *Tamga* in its place, with its price stated in the entry.

S1.16 Sources referred to in this supplement

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